

Geometric constructions of graded and rational Cherednik algebras

Hannah Keese

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Abstract

The aim of this dissertation is to understand the geometric constructions of representations of graded and rational Cherednik algebras given in the paper “Geometric representations of graded and rational Cherednik algebras” by Alexei Oblomkov and Zhiwei Yun [OY14]. More specifically, the aim is to give the constructions of representations of graded Cherednik algebras on the cohomology of homogeneous affine Springer fibers as explicitly as possible, while the results for rational Cherednik algebras are merely stated in the final chapter. For further simplicity in understanding, the twisting of the Weyl group of the reductive group G by automorphisms of the Dynkin diagram for G was left out of the definition of the Cherednik algebras and related constructions.

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Chapter 1

Preliminaries

1.1 Notation

Throughout this dissertation, all varieties are taken over the field of complex numbers $\mathbb{C}$. Let $F = \mathbb{C}((t)) := \mathbb{C}[[t]][\frac{1}{t}]$ be the field of Laurent series and $\mathcal{O}_F = \mathbb{C}[[t]]$ the ring of formal power series.

Let G be a reductive linear algebraic group over $\mathbb{C}$ and fix a maximal torus and Borel subgroup $A \subset B \subset G$.

Let $\mathbb{X}^*(A) = \text{Hom}_{\text{AlgGr}}(A, \mathbb{G}_m)$ be the set of characters associated to the torus A , $\Phi \subset \mathbb{X}^*(A)$ the set of roots determined by A , and Φ_+ the set of positive roots determined by B . Denote the set of coroots by $\Phi^\vee = \mathbb{X}_*(A) = \text{Hom}_{\text{AlgGr}}(\mathbb{G}_m, A)$. Let $\Delta = \{\alpha_1, \dots, \alpha_r\} \subset \Phi_+$ denote the set of simple roots, where r is the rank of G . Let $W = N_G(T)/T$ be the Weyl group associated to A in G , where $N_G(A)$ denotes the normaliser of A in G .

Let $\mathfrak{g}$ denote the Lie algebra of G , and let $\mathfrak{A} = \text{Lie}(A)$ and $\mathfrak{b} = \text{Lie}(B)$. Let $\tilde{G} = G \otimes_{\mathbb{C}} F$ and $\tilde{\mathfrak{g}} = \mathfrak{g} \otimes_{\mathbb{C}} F$, and similarly for $\tilde{A}$.

1.2 Kostant section

Let $\mathfrak{c} = \mathfrak{g} // G := \text{Specmax}(\mathbb{C}[\mathfrak{g}]^G)$ be the GIT quotient of the Lie algebra $\mathfrak{g}$ under the adjoint action of G . It is a well-known theorem of Chevalley that using the notation above:

Theorem (Chevalley). *The restriction morphism of algebras $\mathbb{C}[\mathfrak{g}] \rightarrow \mathbb{C}[\mathfrak{A}]$ induces an isomorphism between the rings of invariants $\mathbb{C}[\mathfrak{g}]^G \rightarrow \mathbb{C}[\mathfrak{A}]^W$.*

Consequently, $\mathfrak{c} \cong \mathfrak{A} // W$. Furthermore, the quotient space $\mathfrak{c}$ is an r -dimensional affine space, with coordinate functions given by fundamental invariant polynomials $f_1, \dots, f_r$, again with r the rank of G .

Given the definition of $\mathfrak{c}$, we therefore have a G -invariant morphism $\chi : \mathfrak{g} \rightarrow \mathfrak{c}$, which induces a morphism, also denoted $\chi : \mathfrak{g} \otimes_{\mathbb{C}} \mathcal{O}_F \rightarrow \mathfrak{c} \otimes_{\mathbb{C}} \mathcal{O}_F$ over $\mathcal{O}_F$.

Given the natural morphism $\mathfrak{c} \otimes_{\mathbb{C}} \mathcal{O}_F \rightarrow \text{Spec}(\mathcal{O}_F)$, let $\mathfrak{c}_F$ denote the fiber over (0) by $\mathfrak{c}_F$, and note that this fiber is isomorphic to $\mathfrak{c} \otimes_{\mathbb{C}} F$.

Definition. Let $\mathfrak{G}$ be a Lie algebra with Lie bracket denoted by $[\ , \]$

1. An element $x \in \mathfrak{G}$ is *semisimple* if $\text{Ad}(x) : \mathfrak{g} \rightarrow \mathfrak{g}$ is a diagonalisable operator.
2. A semisimple element $x \in \mathfrak{G}$ is *regular* if the dimension of its centraliser $C(x) = \{Z \in \mathfrak{G} \mid [x, Z] = 0\}$ is equal to the rank of $\mathfrak{G}$.

Let $\mathfrak{g}^{\text{rs}}$ denote the dense open subset of semisimple regular elements of $\mathfrak{g}$, and let $\mathfrak{c}^{\text{rs}} = \chi^{-1}(\mathfrak{g}^{\text{rs}})$. Then $\mathfrak{c}_F^{\text{rs}} := \mathfrak{c}^{\text{rs}} \otimes_{\mathbb{C}} F$ is an open subscheme of $\mathfrak{c}_F$.

From our choice of maximal torus A and Borel subgroup $B \supset A$, we can define the element $\rho^\vee = \frac{1}{2} \sum_{\alpha \in \Delta} \alpha^\vee \in \mathfrak{t}$, where $\alpha^\vee$ denotes the dual of the simple root $\alpha \in \Delta$. Letting $e = \sum_{\alpha \in \Delta} x_\alpha$, where x_α is a non-zero vector in the root space of $\mathfrak{g}$ corresponding to the root α , define $f \in \mathfrak{g}$ so that $(e, 2\rho^\vee, f)$ is an $\mathfrak{sl}_2$ -triple called the *principal $\mathfrak{sl}_2$ -triple* in $\mathfrak{g}$ (namely these elements generate a subalgebra of $\mathfrak{g}$ isomorphic to $\mathfrak{sl}_2$). Denote the centraliser of f in $\mathfrak{g}$ by $\mathfrak{g}^f$, and define $\mathfrak{s} := e + \mathfrak{g}^f$.

Theorem (Kostant). *The restriction $\chi|_{\mathfrak{s}} : \mathfrak{s} \rightarrow \mathfrak{c}$ is an isomorphism.*

This induces an isomorphism $\chi|_{\mathfrak{s}} \otimes_{\mathbb{C}} \mathcal{O}_F : \mathfrak{s} \otimes_{\mathbb{C}} \mathcal{O}_F \xrightarrow{\cong} \mathfrak{c} \otimes_{\mathbb{C}} \mathcal{O}_F$.

Definition. The *Kostant section* for $\mathfrak{g} \otimes_{\mathbb{C}} \mathcal{O}_F$ is the inverse morphism of χ , denoted

$$\kappa : \mathfrak{c} \otimes_{\mathbb{C}} \mathcal{O}_F \xrightarrow{\cong} \mathfrak{s} \otimes_{\mathbb{C}} \mathcal{O}_F \subset \mathfrak{g} \otimes_{\mathbb{C}} \mathcal{O}_F.$$

1.3 Ind-schemes

From the structure of the infinite-dimensional field-extension $F = \mathbb{C}((t))$ over $\mathbb{C}$, we will often be considering “limits” or infinite unions of schemes. Strictly speaking, these limits will no longer have the geometric structure of a scheme, but will instead have the structure of an “ind-scheme”. We define ind-schemes as seen in [Gör10]. We restrict here to the case of schemes over $\mathbb{C}$, though this can be extended to more general rings.

Definition. A $\mathbb{C}$ -space is a functor $\mathcal{F} : \mathbf{Sch}^{\text{op}} \rightarrow \mathbf{Set}$ that is a sheaf in the fpqc-topology. A *morphism* of $\mathbb{C}$ -spaces is a natural transformation of functors.

Concretely, this means that given a scheme X and an open cover $\{U_i\}$ of X , then the sequence

$$\mathcal{F}(X) \rightarrow \prod_i \mathcal{F}(U_i) \rightrightarrows \prod_{i,j} \mathcal{F}(U_i \cap U_j)$$

is exact, namely that the left arrow is injective and equalises the two right arrows. Furthermore, given any faithfully flat $\mathbb{C}$ -algebra homomorphism $R \rightarrow R'$, the sequence

$$\mathcal{F}(\text{Spec} R) \rightarrow \mathcal{F}(\text{Spec} R') \rightrightarrows \mathcal{F}(\text{Spec} R' \times_{\text{Spec} R} \text{Spec} R')$$

is exact.

Definition. An *ind-scheme* X is a $\mathbb{C}$ -space that is the inductive limit of a direct system of schemes over the natural numbers: $X = \lim_{\rightarrow n} X_n$, where each X_n is a scheme. If the transition functions $X_n \rightarrow X_m$ for $n \leq m$ are all closed immersions, say that X is a *strict ind-scheme*.

Example. Consider the ring of formal power series $\mathbb{C}[[t]]$. This ring is determined by its coefficients and can thus be given the structure of a $\mathbb{C}$ -scheme by viewing it as the set of $\mathbb{C}$ -points of the (infinite) scheme $\mathbb{A}^\infty = \operatorname{Spec}(\mathbb{C}[x_i]_{i \in \mathbb{N}})$. Then $\mathbb{A}^\infty$ is a countable product of affine lines $\mathbb{A}^1$. Similarly the doubly-infinite product of affine lines $\prod_{i \in \mathbb{Z}} \mathbb{A}^1$ is a scheme, given as $\operatorname{Spec}(\mathbb{C}[x_i]_{i \in \mathbb{Z}})$.

However, the field of Laurent series $F = \mathbb{C}((t))$ is not a scheme, since the condition that only finitely many coefficients with negative indices are non-zero cannot be expressed as the zero locus of polynomials. However it is contained in the doubly infinite product $\prod_{i \in \mathbb{Z}} \mathbb{A}^1$, and can clearly be given the structure of an ind-scheme as follows:

$$F = \bigcup_{i \leq 0} t^i \mathbb{C}[[t]]$$

To the reductive linear algebraic group G , associate a group ind-scheme, called the Kac-Moody group that is constructed in the following sections

1.4 The loop group

Given the reductive linear algebraic group G over $\mathbb{C}$, consider the functor LG that to each $\mathbb{C}$ -algebra R associates the $R((t))$ -points of G : $LG(R) = G(R((t)))$. This functor is representable by an ind-scheme, called the *loop group* of G , also denoted LG .

We can similarly consider the functor that to a $\mathbb{C}$ -algebra R associates the $R[[t]]$ -points of G : $L^+G(R) = G(R[[t]])$. This functor is in fact representable by an infinite-dimensional scheme, called the *positive loop group* L^+G .

Lemma. 1. The positive loop group L^+G of a linear algebraic group G is a scheme over $\mathbb{C}$.
2. The loop group LG of a linear algebraic group G is an ind-scheme over $\mathbb{C}$.

1.5 Determinant line bundles

We give here the definition of a line bundle on an ind-scheme, and give an example that will be used in later constructions.

Definition. A *line bundle* on an ind-scheme $X = \cup_{n \in \mathbb{N}} X_n$ is a family of line bundles $\mathcal{L}_n \rightarrow X_n$ that is compatible with the restriction morphisms $X_n \rightarrow X_{n-1}$.

Let R be a $\mathbb{C}$ -algebra and let M be a projective R -module with R -dimension m . Then denote the highest non-zero exterior power $\Lambda^m M$ by $\det(M)$. Consider the adjoint action of the loop group LG on $\Lambda_0 := \mathfrak{g} \otimes_{\mathbb{C}} R((t))$ and define the *relative determinant line* of the $R[[t]]$ -submodule $\Lambda_g := \operatorname{Ad}(g)\Lambda_0$ with respect to Λ_0 to be

$$\det(\Lambda_g : \Lambda_0) = (\det(\Lambda_g / (\Lambda_g \cap \Lambda_0)) \otimes_R (\det(\Lambda_0 / (\Lambda_g \cap \Lambda_0)))^{\otimes -1}$$

The *determinant line bundle* $\det_G$ on the loop group is determined by the functor that for each $\mathbb{C}$ -algebra R sends $g \in LG$ to the relative determinant line $\det(\Lambda_g : \Lambda_0)$, which is an invertible R -module.

The module Λ_g depends only on the coset $g \cdot G(R[[t]])$, so the line bundle $\det_G$ descends to a line bundle on the affine flag variety $Fl_G = LG/L^+G$.

1.6 The associated Kac-Moody group

First construct a central extension of the loop group LG . Let G^{cen} be the total space of the line bundle $\mathcal{L}_{\text{can}} \rightarrow LG$. This is determined by the functor that to any $\mathbb{C}$ -algebra R associates the set

$$G^{\text{cen}}(R) = \{(g, \tau) \mid g \in LG \text{ and } \tau : R \xrightarrow{\sim} \det(\Lambda_g : \Lambda_0) \text{ is an } R\text{-linear trivialisation of } \mathcal{L}_{\text{can}}\}$$

From the group structure on G^{cen} there is an exact sequence of group ind-schemes:

$$1 \rightarrow \mathbb{G}_m^{\text{cen}} \rightarrow G^{\text{cen}} \rightarrow LG \rightarrow 1,$$

where one-dimensional torus $\mathbb{G}_m^{\text{cen}}$ is the fiber of G^{cen} over the identity in the loop group.

Definition. The *affine Kac Moody group* associated to the group G is the extension

$$G_{\text{KM}} := G^{\text{cen}} \rtimes \mathbb{G}_m^{\text{rot}},$$

where $\mathbb{G}_m^{\text{rot}}$ acts on LG by rotation of loops, and acts as the identity on $\mathbb{G}_m^{\text{cen}}$.

Rotation of loops is given by scaling the t -variable in $LG(R((t)))$ by elements of $\mathbb{G}_m^{\text{rot}} \cong \mathbb{C}^\times$: $\lambda \cdot x \otimes f(t) = x \cdot f(\lambda t)$ for all $\lambda \in \mathbb{G}_m^{\text{rot}}$.

The maximal torus $A \subset G$ extends to a torus $A_{\text{KM}} := \mathbb{G}_m^{\text{cen}} \times A \times \mathbb{G}_m^{\text{rot}} \subset G_{\text{KM}}$.

Let $\mathfrak{a} = \mathbb{X}_*(A) \otimes_{\mathbb{Z}} \mathbb{Q}$ and $\mathfrak{a}_{\text{KM}} = \mathbb{X}_*(A_{\text{KM}}) \otimes_{\mathbb{Z}} \mathbb{Q}$, and denote their respective duals by $\mathfrak{a}^*$ and $\mathfrak{a}_{\text{KM}}^*$.

Denote the generators of $\mathbb{X}^*(\mathbb{G}_m^{\text{rot}})$ and $\mathbb{X}^*(\mathbb{G}_m^{\text{cen}})$ by δ and Λ_{can} respectively and denote the generators of $\mathbb{X}_*(\mathbb{G}_m^{\text{rot}})$ and $\mathbb{X}_*(\mathbb{G}_m^{\text{cen}})$ by ∂ and K_{can} respectively.

1.7 The extended affine Weyl group

Let $\Delta = \{\alpha_1, \dots, \alpha_r\}$ be the set of simple roots of the reductive group G and let β be the highest root, with $\{a_0 = 1, a_1, \dots, a_r\} \subset \mathbb{Z}$ such that $\beta = \sum_{i=1}^r a_i \alpha_i$. Let $\alpha_0 = \delta - \beta$. Then $\Delta_{\text{aff}} = \{\alpha_0, \alpha_1, \dots, \alpha_r\}$ is the set of affine roots of G_{KM} .

Given the maximal torus $A \subset G$, this gives rise to a maximal torus $\tilde{A} := A \otimes_{\mathbb{C}} F \subset G \otimes_{\mathbb{C}} F =: \tilde{G}$.

Definition. The *extended affine Weyl group* $\tilde{W}$ is the quotient $\tilde{W} = N_{\tilde{G}} \tilde{A} / (A \otimes_{\mathbb{C}} \mathcal{O}_F)$.

By definition, the Weyl group associated to $\tilde{A} \subset \tilde{G}$ is $W = N_{\tilde{G}} \tilde{A} / \tilde{A}$, so there is a canonical short exact sequence

$$1 \rightarrow T \rightarrow \tilde{W} \rightarrow W \rightarrow 1$$

where $T \cong (A \otimes_{\mathbb{C}} F) / (A \otimes_{\mathbb{C}} \mathcal{O}_F)$

Another presentation of the extended affine Weyl group is obtained by considering the affine Weyl group W_{aff} , the subgroup of $\tilde{W}$ generated by reflections corresponding to the affine roots $\alpha \in \Delta_{\text{aff}}$. Letting $\Omega_I := T/\Lambda$ be the fundamental group of G , where Λ is the coroot lattice of G , there is an exact sequence

$$1 \rightarrow W_{\text{aff}} \rightarrow \tilde{W} \rightarrow \Omega_I \rightarrow 1$$

so that $\tilde{W} = W_{\text{aff}} \rtimes \Omega_I$.

Moreover, Ω_I can be identified with $N_{\tilde{G}}(I)/I$, where I is the chosen Iwahori subgroup of $\tilde{G}$ [OY14].

Definition. The *Killing form* on $\mathbb{X}_*(A)$ is given by

$$B(\cdot, \cdot) : \mathbb{X}_*(A) \times \mathbb{X}_*(A) \rightarrow \mathbb{Z}$$

$$(x, y) \mapsto \sum_{\alpha \in \Phi} \langle \alpha, x \rangle \langle \alpha, y \rangle.$$

where Φ is the set of roots of G .

The Killing form is W -invariant symmetric bilinear form on $\mathbb{X}_*(A)$, and moreover on $\mathfrak{a}$, and can be written as

$$B = \sum_{\alpha \in \Phi} \alpha \otimes \alpha \in \text{Sym}^2(\mathfrak{a})^W.$$

Extend the Killing form to a symmetric bilinear form B_{KM} on $\mathfrak{a}_{\text{KM}}$:

$$B_{\text{KM}} = B + \delta \otimes \Lambda_{\text{can}} + \Lambda_{\text{can}} \otimes \delta$$

Lemma. The symmetric bilinear form B_{KM} is $\widetilde{W}$ -invariant on $\mathfrak{a}_{\text{KM}}$.

1.8 Homogeneous elements

Let $F_n := \mathbb{C}((t^{1/n}))$ for each $n \in \mathbb{N}^*$. Then $\overline{F} := \bigcup_{n \in \mathbb{N}^*} F_n$ is an algebraic closure of F . The one-dimensional torus $\mathbb{G}_m^{\text{rot}}$ acts on $\mathfrak{g} \otimes_{\mathbb{C}} \overline{F}$ by rotation of loops: for all $g \in \mathfrak{g}$, $f(t) \in F$ and $\zeta \in \mathbb{C}^\times \cong \mathbb{G}_m^{\text{rot}}$,

$$\zeta \cdot (g \otimes f(t)) = g \otimes f(\zeta t).$$

The one-dimensional torus $\mathbb{G}_m^{\text{cen}}$ also acts on $\tilde{\mathfrak{g}}$, by dilation: for g and $f(t)$ as above, and for all $\xi \in \mathbb{C}^\times \cong \mathbb{G}_m^{\text{cen}}$,

$$\xi \cdot (g \otimes f(t)) = \xi g \otimes f(t).$$

The action of the two tori can also be defined on $\mathfrak{c} \otimes_{\mathbb{C}} \overline{F}$, so that the quotient map $\chi : \mathfrak{g} \otimes \overline{F} \rightarrow \mathfrak{g} \otimes \overline{F}$ is $\mathbb{G}_m^{\text{dil}} \times \mathbb{G}_m^{\text{rot}}$ -equivariant.

For each rational number $\nu \in \mathbb{Q}$, define a one-dimensional torus $\hat{\mathbb{G}}_m(\nu) \subset \mathbb{G}_m^{\text{dil}} \times \mathbb{G}_m^{\text{rot}}$ via the following:

$$\hat{\mathbb{G}}_m(\nu) = \{(\exp(-z), \exp(\nu z)) \mid z \in \mathbb{C}\} \subset \mathbb{C}_{\text{dil}}^\times \times \mathbb{C}_{\text{rot}}^\times$$

Definition. Let $\nu \in \mathbb{Q}$.

1. An element $a \in \mathfrak{c}^{\text{rs}} \otimes_{\mathbb{C}} \overline{F}$ is *homogeneous of slope ν* if a is fixed under the action of the torus $\hat{\mathbb{G}}_m(\nu)$. Denote the set of all homogeneous elements of slope ν by $\mathfrak{c}(F)_\nu^{\text{rs}}$.
2. A regular semisimple element $\gamma \in \mathfrak{g} \otimes \overline{F}$ is *homogeneous of slope ν* if $\chi(\gamma) \in \mathfrak{c}^{\text{rs}} \otimes \overline{F}$ is.

1.9 The graded Cherednik algebra

As a vector space over $\mathbb{Q}$, the graded rational Cherednik algebra associated to the reductive group $\mathbb{G}$ and the outer automorphism θ ,

$$\mathfrak{H}^{\text{gr}} = \mathfrak{H}^{\text{gr}}(\mathbb{G}, \theta)$$

$$:= \text{Sym}((\mathbb{X}^*(\mathbb{G}_m^{\text{dil}}) \otimes_{\mathbb{Z}} \mathbb{Q}) \oplus \mathfrak{a}_{\text{KM}}^*) \otimes \mathbb{Q}[\widetilde{W}].$$

Thus, letting u be the generator of $\mathbb{X}^*(\mathbb{G}_m^{\text{dil}})$, and recalling that $\mathfrak{a}_{\text{KM}}^* = \mathfrak{a}^* \oplus \mathbb{Q}\delta \oplus \mathbb{Q}\Lambda_{\text{can}}$, where δ is the generator of $\mathbb{X}^*(\mathbb{G}_m^{\text{rot}})$ and Λ_{can} is the generator of $\mathbb{X}^*(\mathbb{G}_m^{\text{cen}})$ then we also have:

$$\mathfrak{H}^{\text{gr}} = \mathbb{Q}[u, \delta] \otimes \text{Sym}(\mathfrak{a}^*) \otimes \mathbb{Q}[\widetilde{W}]$$

with grading defined by the following:

$$\begin{aligned} \deg(\tilde{w}) &= 0 & \text{for all } \tilde{w} \in \widetilde{W} \\ \deg(u) &= 2 \\ \deg(\xi) &= 2 & \text{for all } \xi \in \mathfrak{a}_{\text{KM}}^* \end{aligned}$$

The algebra structure of $\mathfrak{H}^{\text{gr}}$ is given by:

G1 u is central in $\mathfrak{H}^{\text{gr}}$.

G2 $\text{Sym}(\mathfrak{a}_{\text{KM}}^*)$ and $\mathbb{Q}[\widetilde{W}]$ are subalgebras.

G3 For each affine simple reflection s_i in W_{aff} and every $\xi \in \mathfrak{a}_{\text{KM}}^*$,

$$s_i \xi - {}^{s_i} \xi s_i = \langle \xi, \alpha_i^\vee \rangle u.$$

G4 For any $w \in \Omega_I$ and $\xi \in \mathfrak{a}_{\text{KM}}^*$,

$$w\xi = {}^w \xi w$$

By definition, u is central. Furthermore, using relations **G3** and **G4**, δ is also central, hence we consider u and δ as parameters. Note also that by the $\widetilde{W}$ -invariance of B_{KM} , $B_{\text{KM}} \in \text{Sym}^2(\mathfrak{a}_{\text{KM}})^{\widetilde{W}}$ is also central in $\mathfrak{H}^{\text{gr}}$.

For $\nu \in \mathbb{Q}$, let

$$\mathfrak{H}_\nu^{\text{gr}} = \mathfrak{H}^{\text{gr}} / (u + \nu\delta)$$

and denote the image of δ in $\mathfrak{H}_\nu^{\text{gr}}$ by ϵ .

Let $\mathfrak{H}_{\nu, \epsilon=1}^{\text{gr}} := \mathfrak{H}_\nu^{\text{gr}} / (\epsilon - 1)$ be the specialisation of ϵ to 1. This is known as the trigonometric Cherednik algebra. The grading on $\mathfrak{H}^{\text{gr}}$ induces a filtration (not a grading), called the *cohomological filtration*, on $\mathfrak{H}_{\nu, \epsilon=1}^{\text{gr}}$: denote by $\mathfrak{H}_{\nu, \epsilon=1}^{\text{gr}, \leq i} \subset \mathfrak{H}_{\nu, \epsilon=1}^{\text{gr}}$ the image of the degree $\leq i$ part of $\mathfrak{H}_\nu^{\text{gr}}$.

1.10 The rational Cherednik algebra

As a vector space over $\mathbb{Q}$, the bigraded rational Cherednik algebra $\mathfrak{H}^{\text{rat}} = \mathfrak{H}^{\text{rat}}(G)$ is the tensor product

$$\mathfrak{H}^{\text{rat}} = \mathbb{Q}[\delta, u] \otimes \text{Sym}(\mathfrak{a}) \otimes \text{Sym}(\mathfrak{a}^*) \otimes \mathbb{Q}[W] \otimes \mathbb{Q}[\Lambda_{\text{can}}]$$

where the bigrading is given by:

$$\begin{aligned} \deg(\delta) &= (2, 0) \\ \deg(u) &= (2, 0) \\ \deg(\Lambda_{\text{can}}) &= (2, 2) \\ \deg(\eta) &= (0, -1) \quad \text{for all } \eta \in \mathfrak{a} \\ \deg(\xi) &= (2, 1) \quad \text{for all } \xi \in \mathfrak{a}^* \\ \deg(w) &= (0, 0) \quad \text{for all } w \in W \end{aligned}$$

The first component is called the *cohomological grading* and the second component is called the *perverse grading*.

The algebra structure of $\mathfrak{H}^{\text{rat}}$ is given by:

- R1** u and δ are central elements in $\mathfrak{H}^{\text{rat}}$. Λ_{can} commutes with $\mathfrak{a}^*$ and W .
- R2** $\text{Sym}(\mathfrak{a})$, $\text{Sym}(\mathfrak{a}^*)$ and $\mathbb{Q}[W]$ are subalgebras.
- R3** For all $\eta \in \mathfrak{a}$, $\xi \in \mathfrak{a}^*$ and $w \in W$, the following identities hold:

$$w\eta = {}^w\eta w, \quad w\xi = {}^w\xi w$$

- R4** For all $\eta \in \mathfrak{a}$ and all $\xi \in \mathfrak{a}^*$, the following identity holds:

$$[\eta, \xi] = \langle \xi, \eta \rangle \delta + \frac{1}{2} \left(\sum_{\alpha \in \Phi} \langle \xi, \alpha^\vee \rangle \langle \alpha, \eta \rangle s_\alpha \right) u$$

where $s_\alpha \in W$ is the reflection associated to the root α .

- R5** For all $\eta \in \mathfrak{a}$, $[\eta, \Lambda_{\text{can}}] = -\eta^* \in \mathfrak{a}^*$, where η^* is the image of η under the isomorphism $\mathfrak{a} \xrightarrow{\sim} \mathfrak{a}^*$ induced by the Killing form $\mathbb{B}$.

The extension B_{KM} of the Killing form to the Kac-Moody group can be viewed as an element of $\mathfrak{H}^{\text{rat}}$ of bidegree $(4, 2)$. Since B_{KM} is $\widetilde{W}$ -invariant, it is central in $\mathfrak{H}^{\text{rat}}$. For $\nu \in \mathbb{Q}$, define the *rational Cherednik algebra with central charge ν* by:

$$\mathfrak{H}_\nu^{\text{rat}} = \mathfrak{H}^{\text{rat}} / (u + \nu\delta, B_{\text{KM}}).$$

Again denoting the image of $\delta = -u/\nu$ by ϵ , the usual rational Cherednik algebra in the literature is the specialisation of $\mathfrak{H}_\nu^{\text{rat}}$ to $\epsilon = 1$:

$$\mathfrak{H}_{\nu, \epsilon=1}^{\text{rat}} := \mathfrak{H}_\nu^{\text{rat}} / (\epsilon - 1).$$

Then $\mathfrak{H}_{\nu, \epsilon=1}^{\text{rat}}$ has a grading induced by the perverse grading, and a filtration induced by the cohomological grading on $\mathfrak{H}^{\text{rat}}$.

Chapter 2

Geometry

2.1 Affine Grassmannians and affine flag varieties

Recall from the previous section that LG denotes the loop group associated to a linear algebraic group G over $\mathbb{C}$, and L^+G denotes the positive loop group of G .

Definition. The *affine Grassmannian* Gr_G associated to a linear algebraic group G is the quotient $\mathbb{C}$ -space LG/L^+G .

Lemma. *Let G be a linear algebraic group over $\mathbb{C}$. Then the affine Grassmannian Gr_G associated to G is an ind-scheme over $\mathbb{C}$. If G is reductive, then Gr_G is ind-projective.*

Analogously to the finite case, we can construct the affine flag variety associated to a linear algebraic group G . Consider the natural morphism $\pi : L^+G(\mathbb{C}) \rightarrow G(\mathbb{C})$, which sends the variable t to 0 and let $B \subset G$ be a Borel subgroup. Then we call $\pi^{-1}(B) \subset L^+G(\mathbb{C})$ an *Iwahori subgroup* of LG . A *parahoric* subgroup $P \subseteq L^+G(\mathbb{C})$ is similarly an analogue of parabolic subgroups of G . For what follows, we fix an Iwahori subgroup I . We say that a parahoric subgroup P is *standard* if it contains the fixed Iwahori I .

Definition. Let G be a linear algebraic group over $\mathbb{C}$ and let $I \subseteq L^+G$ be an Iwahori subgroup. The *affine flag variety* Fl of G is the quotient $\mathbb{C}$ -space LG/I .

Given a standard parahoric subgroup $P \subset LG$, consider LG as an ind-scheme over $\mathbb{C}$, and P as a group scheme of infinite type over $\mathbb{C}$. Then we can define the *partial affine flag variety* of type P as the fppf quotient $\mathrm{Fl}_P = LG/P$. Observe that when $P = G(\mathcal{O}_F)$, then $\mathrm{Fl}_G = \mathrm{Gr}_G$.

Lemma. *Let G be a linear algebraic group. Then the flag variety Fl_G is an ind-scheme over $\mathbb{C}$.*

2.2 Springer theory

For some basic results and notation for linear algebraic groups, we follow [CG97] and [Bor91]. Let G be a complex reductive connected algebraic group and let $\mathfrak{g} = \mathrm{Lie}(G)$, and note that there is a G -action on $\mathfrak{g}$ by the adjoint action. Fix a maximal torus $A \subseteq G$ and a Borel subgroup B containing A . Let U be the unipotent radical of B , so that $B = A \cdot U$. Let

$\mathfrak{b} = \text{Lie}(B)$, $\mathfrak{a} = \text{Lie}(A)$ and $\mathfrak{n} = \text{Lie}(U)$, so that $\mathfrak{n} = [\mathfrak{b}, \mathfrak{b}]$. Let $\text{rk}(\mathfrak{g}) = \dim(\mathfrak{a})$ be the *rank* of $\mathfrak{g}$ (and note that this does not depend on the choice of A). Let W be the associated Weyl group.

Let $\tilde{\mathfrak{g}} = G \times^B \mathfrak{b}$, the quotient of $G \times \mathfrak{b}$ by the action of B given by: $b \cdot (g, x) = (g \cdot b^{-1}, \text{Ad}(b)(x))$ for all $g \in G, b \in B, x \in \mathfrak{b}$. Note that this action is free since its action on the first factor of $G \times \mathfrak{b}$ is free.

Theorem (Grothendieck). *The following diagram*

$$\begin{array}{ccc}
 & \tilde{\mathfrak{g}} & \\
 \nu \swarrow & & \searrow \phi \\
 \mathfrak{a} & & \mathfrak{g} \\
 \psi \searrow & & \swarrow \chi \\
 & \mathfrak{a}/W &
 \end{array} \tag{2.1}$$

where

1. $\nu : \tilde{\mathfrak{g}} \rightarrow \mathfrak{a}$ sends a pair $(g, x) \in G \times \mathfrak{b}$ to the projection of $x \in \mathfrak{b} = \mathfrak{a} \oplus \mathfrak{n}$ to its $\mathfrak{a}$ -component.
2. $\phi : \tilde{\mathfrak{g}} \rightarrow \mathfrak{g}$ sends a pair $(g, x) \in G \times \mathfrak{b}$ to $\text{Ad}(g)(x)$
3. $\psi : \mathfrak{a} \rightarrow \mathfrak{a}/W$ is the canonical projection, and
4. $\chi : \mathfrak{g} \rightarrow \mathfrak{a}/W$ is induced by the algebra homomorphism $\mathbb{C}[\mathfrak{a}/W] := \mathbb{C}[\mathfrak{a}]^W \cong \mathbb{C}[\mathfrak{g}]^G \hookrightarrow \mathbb{C}[\mathfrak{g}]$

is a simultaneous resolution of the map χ .

Concretely, this says that:

1. The morphism ϕ is proper.
2. The morphism ν is smooth.
3. The morphism ψ is finite.
4. For each $a \in \mathfrak{a}$, the restriction $\phi|_{\nu^{-1}(a)} : \nu^{-1}(a) \rightarrow \chi^{-1}(\psi(a))$ is a resolution of singularities.

Definition. Let $x \in \mathfrak{g}$. The *affine Springer fiber* at x is the fiber of ϕ in (2.1) above, denoted Sp_x .

This is a reduced scheme, that can be described as follows: $\text{Sp}_x = \{g \cdot B \mid \text{Ad}(g^{-1})(x) \in \mathfrak{b}\}$.

2.3 Affine Springer fibers

We extend the notion of Springer fibers to the affine case, following the definition of Kazhdan and Lusztig in [KL88]. As before let G be a reductive group over $\mathbb{C}$, denote its Lie algebra by $\mathfrak{g}$, and write $\mathfrak{g}(F) := \mathfrak{g} \otimes_{\mathbb{C}} F$. We denote Iwahori subgroups by I and parahoric subgroups by P .

Definition. Let $\gamma \in \mathfrak{g}(F)$ be regular semisimple. The *affine Springer fiber* of γ of type P is the reduced closed ind-subscheme $\mathrm{Sp}_{P,\gamma} := \{g \cdot P \in \mathrm{Fl}_P \mid \mathrm{Ad}(g^{-1})(\gamma) \in \mathrm{Lie}(P)\}$ of the partial flag variety Fl_P . If $P = I$, we simply write Sp_γ .

The requirement that γ be regular semisimple ensures that $\dim(\mathrm{Sp}_{P,\gamma}) < \infty$, though it is possibly an infinite union of finite-dimensional projective varieties.

Supposing that $a \in \mathfrak{c}(F)_\nu^{\mathrm{rs}}$ is homogeneous of slope $\nu > 0$, and let $\gamma = \kappa(a) \in \mathfrak{g}$. Denote the Springer fiber $\mathrm{Sp}_{P,\gamma}$ by $\mathrm{Sp}_{P,a}$, and say that this Springer fiber is *homogeneous*.

We define families of homogeneous affine Springer fibers with equal slope as follows: let

$$\tilde{q}_{P,\nu} : \widetilde{\mathrm{Sp}}_{P,\nu} \rightarrow \mathfrak{g}(F)_\nu^{\mathrm{rs}}$$

be the family of affine Springer fibers over $\mathfrak{g}(F)_\nu^{\mathrm{rs}}$ in Fl_P , and let

$$q_{P,\nu} : \mathrm{Sp}_{P,\nu} \rightarrow \mathfrak{c}(F)_\nu^{\mathrm{rs}}$$

be the restriction to $\mathfrak{c}(F)_\nu^{\mathrm{rs}}$ via the Kostant section $\kappa : \mathfrak{c}(F)_\nu \hookrightarrow \mathfrak{g}(F)_\nu$ so that for each $a \in \mathfrak{c}(F)_\nu^{\mathrm{rs}}$, the Springer fiber $\mathrm{Sp}_{P,a}$ is the fiber of $q_{P,\nu}$ at a .

2.4 Principal G -bundles

2.4.1 Topological principal G -bundles

First, consider the more general and intuitive topological case. Let G be a topological group. Let X be a topological space. In the next section we consider more specifically the algebraic case. Further details can be found in [Mit11].

Definition. A *right G -action* of the topological group G on a topological space X is a map $\Phi : X \times G \rightarrow X$ denoted $x \cdot g := \Phi(x, g)$, such that for all $x \in X$ $x \cdot e = x$, where e is the identity element of G , and $x \cdot (gh) = (x \cdot g) \cdot h$ for all $g, h \in G$.

A *right G -space* is a topological space X together with a continuous right G -action.

A map $\phi : X \rightarrow Y$ of left G -spaces is G -equivariant if for all $g \in G$ and all $x \in X$, $\phi(x \cdot g) = \phi(x) \cdot g$.

Naturally, there is an analogous definition for left G -actions and G -spaces.

Definition. Let B be a topological space. A *principal G -bundle* is a pair (P, π) , where P is a right G -space, and $\pi : P \rightarrow B$ is a G -equivariant map, such that the action of G on B is trivial, and π satisfies the following local triviality condition:

There exists an open cover $\{U_i\}_{i \in I}$ of B and G -equivariant homeomorphisms $\phi_i : \pi^{-1}(U_i) \rightarrow U_i \times G$ such that the following diagram commutes:

$$\begin{array}{ccc} \pi^{-1}(U_i) & \xrightarrow{\phi_i} & U_i \times G \\ \downarrow \pi & \swarrow p_1 & \\ U & & \end{array}$$

where $U_i \times G$ is a right G -space with action given by $(u, g) \cdot h = (u, gh)$ for all $u \in U_i, g, h \in G$ and p_1 is the projection onto the first factor.

As a consequence of the local triviality condition, G acts freely on P :

Let $x \in X$ and suppose $g \in G$ such that $x \cdot g = x$. Let U_i be an open neighbourhood of $\pi(x)$ in B satisfying the local triviality condition. Then $\phi_i(x \cdot g) = \phi_i(x) \cdot g = (\pi(x), g') \cdot g$ for some

$$\begin{array}{ccc} \phi_i(x) & \xlongequal{\quad} & (\pi(x), g'g) \\ g' \in G. \text{ Thus, } & \parallel & \parallel \quad \text{and } g = e. \\ \phi_i(x) & \xlongequal{\quad} & (\pi(x), g') \end{array}$$

Therefore π factors through the homeomorphism $\bar{\pi} : P/G \xrightarrow{\sim} B$, and B can be seen as the orbit space of P .

Definition. A *morphism* of principal G -bundles $(P, \pi_P), (Q, \pi_Q)$ over B is a G -equivariant map $\sigma : P \rightarrow Q$ such that $\pi_Q \circ \sigma = \pi_P$.

A G -bundle is said to be *trivial* if it is isomorphic to the product bundle $B \times G \rightarrow G$.

Proposition. Any morphism of principal G -bundles is an isomorphism.

Proof. Let $\sigma : P \rightarrow Q$ be a morphism of G -bundles over B . Suppose first that P and Q are trivial. Then by G -equivariance of σ , $\sigma(b, g) = (b, f(b)g)$ where $f : B \rightarrow G$ is a continuous map. Then σ has an inverse, given by $(b, g) \mapsto (b, f(b)^{-1}g)$.

All G -bundles are locally trivial and we can clearly glue these isomorphisms, hence all morphisms are isomorphisms. $\square$

As a consequence, the category of principal G -bundles over a space B is a groupoid.

2.4.2 Algebraic principal G -bundles

Here, we follow [EG98] Let G be a linear algebraic group acting on a scheme X . Denote the action map by:

$$\begin{aligned} A : G \times X &\rightarrow X \times X \\ (g, x) &\mapsto (g \cdot x, x) \end{aligned}$$

Let $\Delta_X = \{(x, x) \in X \times X \mid x \in X\}$ be the diagonal in $X \times X$. Given a group action $A : G \times X \rightarrow X \times X$, say that the restriction $A|_{A^{-1}(\Delta_X)} : A^{-1}(\Delta_X) \rightarrow \Delta_X$ is the stabiliser group scheme of X . It is clear that the fiber of the restriction above a point x_0 is the stabiliser G_{x_0} of x_0 :

$$(A|_{A^{-1}(\Delta_X)})^{-1}(x_0) = \{(g, x) \in G \times X \mid x = x_0 \text{ and } g \cdot x_0 = x_0\} \cong \{g \in G \mid g \cdot x_0 = x_0\} = G_{x_0}$$

Definition. Let G and X be as above. The group action of G on X is:

- *closed* if the orbits of geometric points are closed in X .
- *free* if the action map A is a closed embedding.
- *set-theoretically free* if the stabiliser of every point is trivial.

Let $f : X \rightarrow Y$ be a morphism of schemes, and let $X \times_Y X$ be the fiber product of X with itself over Y given by f :

$$\begin{array}{ccc}
X \times_Y X & \xrightarrow{p_2} & X \\
\downarrow p_1 & & \downarrow f \\
X & \xrightarrow{f} & Y
\end{array}$$

Suppose that f is a G -equivariant morphism of schemes equipped with G -actions, and suppose further that the action on Y is trivial. Then $f(g \cdot c) = g \cdot f(x) = f(x)$ for all $x \in X$ and all $g \in G$, and the action map $A_X : G \times X \rightarrow X \times X$ factorises through $X \times_Y X$ by the universal property of the fiber product, hence there is a map $A' : G \times X \rightarrow X \times_Y X$.

Definition. Let G be a linear algebraic group acting on schemes X and Y . Let $f : X \rightarrow Y$ be a flat, surjective, G -equivariant morphism of schemes. Then f is a *principal G -bundle* (or *G -torsor*) if the action of G is trivial on Y , and the map $A' : G \times X \rightarrow X \times_Y X$ described above is an isomorphism.

Remark. To see that this is similar to the definition of a principal G -bundle in the topological setting above, it can be shown that the second condition is equivalent to the condition that the morphism f is locally trivial in the étale topology, i.e. that there is an étale cover $U \rightarrow Y$ such that $U \times_Y X \cong U \times G$.

Proposition ([EG98], Proposition 23). *Let G be a linear algebraic group, let U be a scheme equipped with a free G -action and suppose that a principal bundle quotient $U \rightarrow U/G$ exists in the category of schemes. Suppose X is a quasi-projective scheme with a linearized G -action. Then a principal bundle quotient $X \times U \rightarrow X \times^G U$ exists in the category of schemes.*

This in particular says that the spaces $G \times^B \mathfrak{b}$ are schemes.

2.5 Sheaves

2.5.1 Pullbacks of sheaves

Definition. Let X and Y be topological spaces, $f : Y \rightarrow X$ a continuous map and $\mathcal{F}$ a sheaf on X . The *pullback* of the sheaf $\mathcal{F}$ by f , denoted $f^{-1}\mathcal{F}$, is the sheaf on Y associated to the presheaf defined by:

$$U \mapsto \lim_{V \supseteq f(U)} \mathcal{F}(V),$$

for U an open set in Y , and the colimit is taken over all open sets V containing $f(U)$.

We further consider pullbacks of sheaves of modules on schemes. For a scheme X , let $\mathcal{O}_X$ denote the structure sheaf of X .

Definition. Let $f : Y \rightarrow X$ be a morphism of schemes, and let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$ -modules. The *pullback* (or *inverse image*) of $\mathcal{F}$ is the sheaf of $\mathcal{O}_Y$ -modules on Y :

$$f^*\mathcal{F} := \mathcal{O}_Y \otimes_{f^{-1}\mathcal{O}_X} f^{-1}\mathcal{F}.$$

Let $\mathcal{G}$ be a sheaf of $\mathcal{O}_Y$ -modules. The *pushforward* (or *direct image*) of $\mathcal{G}$ is the sheaf $f_*\mathcal{G}$ on X defined by:

$$U \mapsto \mathcal{G}(f^{-1}(U))$$

where U is an open set in X . This is a sheaf of $f_*\mathcal{O}_X$ -modules, which inherits the structure of an $\mathcal{O}_Y$ -module via the morphism $f^\# : \mathcal{O}_Y \rightarrow f_*\mathcal{O}_X$ of sheaves on Y defined by the morphism f .

The pullback and pushforward satisfy the following:

Lemma. 1. *The pullback and pushforward are functors of sheaves of $\mathcal{O}$ -modules.*

2. *In particular, for suitably composable morphisms f and g , $(gf)^* = f^*g^*$ and $(gf)_* = g_*f_*$.*

3. *Given a morphism $f : (Y, \mathcal{O}_Y) \rightarrow (X, \mathcal{O}_X)$ of schemes, the functors f_* and f^* form an adjoint pair (f^*, f_*) , namely for any $\mathcal{O}_X$ -module $\mathcal{F}$ and any $\mathcal{O}_Y$ -module $\mathcal{G}$ we have a natural group isomorphism:*

$$\mathrm{Hom}_{\mathcal{O}_Y}(f^*\mathcal{F}, \mathcal{G}) \cong \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{F}, f_*\mathcal{G})$$

4. *Let R be a ring and let R_Z be the constant sheaf with stalk R on Z for any topological space Z and $f : Y \rightarrow X$ a morphism of schemes. Then $f^*(R_X) = R_Y$.*

5. *The pushforward functor is left exact. Denote by $R^i f_*$ its right derived functors.*

2.5.2 Equivariant sheaves

Let G be an algebraic group acting on a variety X . There are morphisms: $G \times X \xrightarrow[p]{a} X$, where $a(g, x) = g \cdot x$ is given by the action of G on X and $p(g, x) = x$ is the projection onto the X factor. Furthermore, let

$$\begin{aligned} m : G \times G &\rightarrow G \\ (g, h) &\mapsto gh \\ p_{23} : G \times G \times X &\rightarrow G \times X \\ (g, h, x) &\mapsto (h, x) \end{aligned}$$

be the group multiplication morphism and a projection along the first G factor, respectively. Then the following diagrams commute:

$$\begin{array}{ccccc} G \times G \times X & \xrightarrow{m \times \mathrm{id}_X} & G \times X & G \times G \times X & \xrightarrow{m \times \mathrm{id}_X} & G \times X & G \times G \times X & \xrightarrow{\mathrm{id}_G \times a} & G \times X \\ \mathrm{id}_G \times a \downarrow & & \downarrow a & p_{23} \downarrow & & \downarrow p & p_{23} \downarrow & & \downarrow p \\ G \times X & \xrightarrow{a} & X & G \times X & \xrightarrow{p} & X & G \times X & \xrightarrow{a} & X \end{array}$$

Definition. Let G be an algebraic group acting on a scheme X . A sheaf $\mathcal{F}$ of $\mathcal{O}_X$ -modules on X is *G -equivariant* if the following conditions hold:

1. There is a given isomorphism of sheaves on $G \times X$

$$\Theta : a^*\mathcal{F} \xrightarrow{\sim} p^*\mathcal{F}.$$

2. The following diagram of sheaves on $G \times G \times X$ commutes:

$$\begin{array}{ccc} (\mathrm{id}_G \times a)^* a^* \mathcal{F} & \xlongequal{\quad} & (m \times \mathrm{id}_X)^* a^* \mathcal{F} \\ \downarrow (\mathrm{id}_G \times a)^* \Theta & & \downarrow (m \times \mathrm{id}_X)^* \Theta \\ (\mathrm{id}_G \times a)^* p^* \mathcal{F} & \xrightarrow{p_{23}^* \Theta} & (m \times \mathrm{id}_X)^* p^* \mathcal{F} \end{array}$$

Namely, the identity $p_{23}^* \Theta \circ (\mathrm{id}_G \times a)^* \Theta = (m \times \mathrm{id}_X)^* \Theta$ holds.

As a consequence, for e the unit of the group G , the following identity holds on the restriction to $e \times X$:

$$\Theta_{e \times X} = \text{id}_{\mathcal{F}} : \mathcal{F} = a^* \mathcal{F}|_{e \times X} \xrightarrow{\simeq} p^* \mathcal{F}|_{e \times X} = \mathcal{F}$$

A *morphism* of G -equivariant sheaves is a morphism of sheaves that commutes with the isomorphism Θ , hence the G -equivariant morphisms on a G -space X form a category, denoted $\mathbf{Sh}_G(X)$.

2.5.3 The derived category of complexes of sheaves

Let X be a scheme over $\mathbb{C}$, and let $\mathbf{Sh}(X)$ denote the category of sheaves on X . Let $D(\mathbf{Sh}(X))$ and $D^b(\mathbf{Sh}(X))$ be the derived and bounded derived category of finite complexes of sheaves on X respectively. A sheaf $\mathcal{F}$ on X is *constructible* if there is an algebraic stratification of $X = \sqcup_{\alpha} X_{\alpha}$ such that for each α , the restriction of $\mathcal{F}$ to X_{α} is a locally constant sheaf of finite-dimensional vector spaces. An object $F \in D^b(\mathbf{Sh}(X))$ is a *constructible complex* if there all its cohomology sheaves $H^i(F)$ are constructible. Denote the full subcategory of constructible complexes in $D^b(\mathbf{Sh}(X))$ by $D^b(X)$. We refer the reader to [KS90] for further details and constructions.

2.5.4 Verdier duality

(As seen in [CG97])

Let X be a topological manifold (in this context, X is generally a scheme over $\mathbb{C}$ with the transcendent or étale topology). Let $i : X \hookrightarrow M$ be a closed embedding of X into a smooth manifold M . Then define a functor $i^! : \mathbf{Sh}(M) \rightarrow \mathbf{Sh}(X)$ by taking germs of sections with support in X : if $\mathcal{F}$ is a sheaf on M , then for each $x \in X$, the stalk of $i^! \mathcal{F}$ at x is:

$$(i^! \mathcal{F})|_x = \lim_{U \ni x} \Gamma_X(U, \mathcal{F})$$

where the direct limit is taken over all open neighbourhoods U of x in M and $\Gamma_X(U, \mathcal{F})$ is the set of sections of $\mathcal{F}$ over U whose support is contained in $X \cap U$.

The functor $i^!$ is left exact. Denote its right derived functor by $Ri^! : D^+(\mathbf{Sh}(M)) \rightarrow D^+(\mathbf{Sh}(X))$. Letting X and M be algebraic varieties, then $Ri^! : D^b(X) \rightarrow D^b(M)$. From now on, we consistently work with derived functors, and hence to simplify notation, we simply write, for example, $i^!$, rather than $Ri^!$.

For any topological manifold X , denote the constant sheaf determined by $\mathbb{C}$ by $\mathbb{C}_X$, and view it as an element of $D^b(X)$ that is concentrated in degree zero.

Definition. The *dualising complex* D_X of an algebraic variety X is the element of $D^b(X)$ given by:

$$D_X = Ri^!(\mathbb{C}_M)[2\dim_{\mathbb{C}} M]$$

Lemma. 1. The dualising complex D_X does not depend on the embedding $i : X \hookrightarrow M$.

2. Let $i : N \hookrightarrow M$ be a closed embedding of a smooth complex variety N into a smooth complex variety M . Then

$$Ri^!(\mathbb{C}_M) = \mathbb{C}_N[-2d] \quad \text{where } d = \dim_{\mathbb{C}} M - \dim_{\mathbb{C}} N$$

3. In particular, if X is smooth, then

$$D_X = \mathbb{C}_X[2\dim_{\mathbb{C}} X]$$

Definition. Let X be a scheme over X , let $\mathcal{F}$ be an object in $D^b(X)$ and let j be any integer. The j -th hyper-cohomology group $H^j(\mathcal{F}) = H^j(X, \mathcal{F})$ of $\mathcal{F}$ is the j -th derived functor of the global sections functor $\Gamma : \mathbf{Sh}(X) \rightarrow \mathbf{Vect}_{\mathbb{C}}$. We make an analogous definition $H_c^j(\mathcal{F})$ for hyper-cohomology with compact support.

Lemma. Let X be a locally compact, finite-dimensional space. Then $H^i(X) = H^i(X, \mathbb{C}_X)$ and $H_i(X) = H^{-i}(X, D_X)$, where the right hand side of each equation is the hypercohomology of the respective complexes over X .

Definition. Let A and B be complexes in $D^b(X)$. The *Ext-groups* in $D^b(X)$ are:

$$\mathrm{Ext}_{D^b(X)}^k(A, B) = \mathrm{Hom}_{D^b(X)}(A, B[k])$$

for each k in $\mathbb{Z}$.

The *internal Hom-complex* $\mathcal{H}om(A, B) \in D^b(X)$ is given by:

$$\mathrm{Ext}_{D^b(X)}^k(A, B) = H^k(X, \mathcal{H}om(A, B))$$

Definition. The *Verdier duality functor* is the contravariant functor on $D^b(X)$ that to each object A associates the object $A^\vee := \mathcal{H}om(A, D_X)$.

The Verdier dual has the following properties:

Lemma. 1. The Verdier dual of the constant sheaf $\mathbb{C}_X$ is the dualising complex D_X .

2. Let $\mathcal{F}$ be an object of $D^b(X)$. Then $(\mathcal{F}[n])^\vee = (\mathcal{F}^\vee)[n]$ and $(\mathcal{F}^\vee)^\vee = \mathcal{F}$.

Let $f : X \rightarrow Y$ be an algebraic map. Then there are (derived) functors:

$$f_* : D^b(X) \rightarrow D^b(Y) \quad \text{and} \quad f^* : D^b(Y) \rightarrow D^b(X),$$

the derived functors of the sheaf-theoretic pushforward and pullback functors, respectively. Verdier duality gives another pair of (derived) functors:

$$f_! : D^b(X) \rightarrow D^b(Y) \quad \text{and} \quad f^! : D^b(Y) \rightarrow D^b(X)$$

given by

$$f_!(\mathcal{F}) = (f_*(\mathcal{F}^\vee))^\vee \quad \text{and} \quad f^!(\mathcal{G}) = (f^*(\mathcal{G}^\vee))^\vee$$

for any complex $\mathcal{F}$ in $D^b(X)$ and any complex $\mathcal{G}$ in $D^b(Y)$.

Lemma. Using the notation above,

1. The pairs (f_*, f^*) and $(f_!, f^!)$ are adjoint pairs of functors. In particular, there are canonical isomorphisms:

$$\mathrm{Hom}(f^*\mathcal{F}, \mathcal{G}) \cong \mathrm{Hom}(\mathcal{F}, f_*\mathcal{G}) \quad \text{and} \quad \mathrm{Hom}(\mathcal{F}, f^!\mathcal{G}) \cong \mathrm{Hom}(f_!\mathcal{F}, \mathcal{G})$$

2. On hyper-cohomology and hypercohomology with compact support respectively, there are isomorphisms

$$H^\bullet(Y, f_*\mathcal{F}) \cong H^\bullet(X, \mathcal{F}) \quad \text{and} \quad \text{Hom}_c^\bullet(Y, f_!\mathcal{F}) \cong H_c^\bullet(X, \mathcal{F})$$

3. On complexes, the following isomorphisms hold:

$$f^*\mathbb{C}_Y \cong \mathbb{C}_X \quad \text{and} \quad f^!D_Y = D_X$$

4. Furthermore, if f is a proper morphism, then $f_! = f_*$ and if f is flat with smooth fibers of complex dimension d , then $f^! = f^*[2d]$.

2.6 Quotient stacks

While we do not attempt to understand stacks in depth, or to give formal definitions, we use certain properties of quotient stacks without proof throughout section (3).

Our motivation for considering stacks is in the definition of a quotient of certain schemes by group actions. While given a group action on a topological space the quotient space is again a topological space, this is unfortunately not necessarily true in the case of schemes. Both the use of algebraic stacks and GIT quotients gives us some alternative (more general) objects to rectify this issue.

Recall that a groupoid is a category in which all morphisms are isomorphisms. In the following, we implicitly work with certain Grothendieck topologies, particularly the étale or fppf topologies, and how this will affect the gluing axioms in the definition of a sheaf. For instance, a covering in this case will be as defined above in the discussion on Grothendieck topologies, and in this topology the intersection of two open sets forming a covering of a scheme X is given by the fibered product $U \times_X V$ and restrictions are given by pullbacks.

As a guide, we use the following informal definition, as in [Hei10], though there are some further details and considerations to be taken into account for a true definition.

Definition. A *stack* is a sheaf of groupoids:

$$\mathcal{M} : \mathbf{Sch} \rightarrow \mathbf{Groupoid}$$

The main example of a stack that interests us here is that of a quotient stack. Let X be a scheme and let G be an algebraic group acting on X .

Definition. The quotient stack $[X/G]$ is the contravariant functor that to each scheme S associates the category $[X/G](S)$ of triples (P, p, f)

$$\begin{array}{ccc} P & \xrightarrow{f} & X \\ \downarrow p & & \\ S & & \end{array} \tag{2.2}$$

such that $p : P \rightarrow S$ is a principal G -bundle and $f : P \rightarrow X$ is a G -equivariant morphism. Morphisms in this category are (iso)morphisms of principal G -bundles that commute with the morphism to X .

The following lemma is clear from the above definition of a quotient stack:

Lemma. *Let X and Y be G -spaces, and let $f : X \rightarrow Y$ be a G -equivariant map. Then f induces a morphism of stacks:*

$$f_G : [X/G] \rightarrow [Y/G]$$

We associate a stack $\underline{S}$ to a scheme S as follows:

Definition. Let S be a scheme. Then the stack $\underline{S}$ associates to any scheme T the set $\underline{S}(T) = \text{Mor}_{\text{Sch}}(T, S)$ considered as a category where the only morphisms are identities, and to any morphism of schemes $f : T \rightarrow R$ associates the functor $\underline{S}(R) \rightarrow \underline{S}(T)$ given by composition with f .

Lemma. (Yoneda lemma for stacks) *Let $\mathcal{M}$ be a stack. Then for any scheme S there is a natural equivalence of categories:*

$$\text{Mor}_{\text{Stack}}(\underline{S}, \mathcal{M}) \cong \mathcal{M}(S)$$

Proof. There is a natural functor $\phi : \text{Mor}_{\text{Stack}}(\underline{S}, \mathcal{M}) \rightarrow \mathcal{M}(S)$ sending a stack morphism $F : \underline{S} \rightarrow \mathcal{M}$ to the object $F(id_S)$ of $\mathcal{M}(S)$.

Conversely, let $\mathcal{E}$ be an object of $\mathcal{M}(S)$. Then for any scheme T and any scheme morphism $f : T \rightarrow S$ define the morphism $\psi(\mathcal{E}) : \underline{S} \rightarrow \mathcal{M}$ by $\psi(\mathcal{E})(f) = f^*(\mathcal{E})$, which is an element of $\mathcal{M}(T)$

The functors ϕ and ψ defined above are inverses:

Let $\mathcal{E}$ be an object of $\mathcal{M}(S)$. Then

$$\phi(\psi(\mathcal{E})) = \phi(\psi(\mathcal{E})) = \psi(\mathcal{E})(id_S) = id_S^*(\mathcal{E}) = \mathcal{E}.$$

Conversely, let $F : \underline{S} \rightarrow \mathcal{M}$, then $\psi(\phi(F)) = \psi(F(id_S))$. Let $f : T \rightarrow S$ be any morphism of schemes, then $\psi(F(id_S))(f) = f^*(F(id_S)) \cong F(f)$, and this isomorphism is natural by functoriality. $\square$

Thus we may identify a scheme S with the stack $\underline{S}$, and in particular, in the case of quotient stacks a morphism of stacks from a scheme is exactly a diagram of the form (2.2). Concretely, the definition of a morphism of stacks from a scheme S to the quotient stack $[X/G]$ is a triple (P, f, g) , where $g : P \rightarrow S$ is a G -bundle and $f : P \rightarrow X$ is a G -equivariant map.

In later constructions, it will be necessary to consider the derived category of sheaves of certain quotients, which is here defined as follows:

Definition. Let G be an algebraic group acting on a scheme X and let $[X/G]$ be the quotient stack of X by G . The *bounded derived category of sheaves of $\mathbb{C}$ -vector spaces* on the stack $[X/G]$ is the bounded derived category of G -equivariant sheaves on X :

$$D^b([X/G]) := D_G^b(X)$$

We also want to show the following (the analogous result on quotient spaces is found in [BL94], section 2.6, and extended to quotient stacks by the definition above):

Proposition (Induction equivalence). *Let G be an algebraic group, and H a closed subgroup of G . Let X be a scheme with an H -action. Then the canonical morphism $\nu : X \rightarrow X \times^H G$ given by $x \mapsto (x, e)$ induces an isomorphism of stacks:*

$$\text{ind} : [X \times^H G/G] \xrightarrow{\sim} [X/H]$$

and furthermore the induced inverse image functor $\nu^ : D_G^b(X \times^H G) \rightarrow D_H^b(X)$ is an equivalence of triangulated categories.*

For what follows, let X and Y be topological spaces, let $\phi : H \rightarrow G$ be a homomorphism of topological groups, and suppose H acts on X and G acts on Y .

- Definition.**
1. A continuous map $f : X \rightarrow Y$ is a ϕ -map if for all $h \in H$ and all $x \in X$, $f(h \cdot x) = \phi(h)f(x)$.
 2. A G -space Y is *free* if G acts freely on Y ($g \cdot x = x$ if and only if $g = e$) and the quotient map $q : Y \rightarrow Y/G$, where Y/G denotes the space of G -orbits of Y , is a locally trivial fibration with fiber G .

2.7 (Co)homology

2.7.1 Borel Moore homology

Throughout the constructions in section 3, we generally use Borel-Moore homology. In the case of compact topological spaces, this is equivalent to ordinary singular homology, though has further good properties in the case of complex algebraic varieties, particularly when these are not necessarily smooth or compact.

Definition. Let X be a topological space. The following are equivalent definitions of Borel-Moore homology $H_*^{\text{BM}}(X)$:

1. Let $\overline{X}$ be a compactification of X such that $(\overline{X}, \overline{X} \setminus X)$ is a CW-pair. Then

$$H_*^{\text{BM}}(X) = H_*(\overline{X}, \overline{X} \setminus X)$$

where $H_*(Y, Z)$ denotes ordinary relative homology for a CW-pair (Y, Z) .

2. Let $C_\bullet^{\text{BM}}(X)$ be the chain complex of infinite singular chains $\sum_{i=0}^\infty a_i \sigma_i$, for σ_i a singular complex, and $a_i \in \mathbb{C}$ such that for each compact set $D \subset X$, there are only finitely many $a_i \neq 0$ such that $D \cap \text{supp} \sigma_i \neq \emptyset$, together with the usual boundary map on singular chains ∂ . Then

$$H_*^{\text{BM}}(X) = H_*(C_\bullet^{\text{BM}}(X), \partial).$$

3. **Poincaré Duality:** Let M be a smooth, oriented manifold of real dimension m such that $X \subset M$ is closed, with a closed neighbourhood U such that X is a proper deformation retract of U . Then

$$H_i^{\text{BM}}(X) \cong H^{m-i}(M, M \setminus X)$$

where on both sides, we take complex coefficients and the right hand side is the ordinary relative singular cohomology.

In particular, for X a smooth oriented (not necessarily compact) manifold of real dimension m , there is a canonical isomorphism given by the orientation of X :

$$H_i^{\text{BM}}(X) \cong H^{m-i}(X)$$

Notation: From now on, we denote the Borel-Moore homology by H_* , and ordinary singular homology by H_*^{ord} .

Borel-Moore homology has the following properties:

Proposition. 1. **Proper pushforward** Let $f : X \rightarrow Y$ be a proper map. Then there is a pushforward map on Borel-Moore homology:

$$f_* H_*(X) \rightarrow H_*(Y).$$

2. **Pullback along open embeddings** Let $g : U \hookrightarrow X$ be an open embedding. Then there is a natural restriction morphism on Borel-Moore homology

$$g^* : H_*(X) \rightarrow H_*(U).$$

3. **The long exact sequence on Borel-Moore homology** Let $i : F \hookrightarrow X$ be a closed subset, and $j : U = X \setminus F \hookrightarrow X$ be the complement in X . Then the diagram

$$F \xhookrightarrow{i} X \xleftarrow{j} U$$

induces a long exact sequence on Borel-Moore homology:

$$\dots \rightarrow H_{k+1}(U) \rightarrow H_k(F) \rightarrow H_k(X) \rightarrow H_k(U) \rightarrow H_{k-1}(F) \rightarrow \dots$$

4. **Fundamental class** Let X be a complex algebraic variety, that is not necessarily smooth or compact, of real dimension m . Then X has a fundamental class $[X] \in H_m(X)$. More precisely, if $X = \cup_{i=1}^n X_i$, where the X_i are the irreducible components of X , then the fundamental class of X is $[X] = \sum_{i=1}^n [X_i] \in H_m(X)$, where $[X_i]$ is the fundamental class of the irreducible component $[X_i]$.

Proof. 1. Let $f : X \rightarrow Y$ be proper. Then let $\hat{X} = X \cup \{\infty\}$ and $\hat{Y} = Y \cup \{\infty\}$ be the one-point compactifications of X and Y respectively, and extend f to $\hat{f} : \hat{X} \rightarrow \hat{Y}$ by setting $\hat{f}(\infty) = \infty$. Then $\hat{f}$ is continuous since f is proper. Then in relative singular homology, there is a pushforward

$$\hat{f}_* : H_*(\hat{X}, \{pt\}) \rightarrow (H_*(\hat{Y}, \{pt\}))$$

which gives the pushforward on Borel-Moore homology.

2. Let $\overline{X}$ be a compactification of X such that $(\overline{X}, \overline{X} \setminus X)$ is a CW-pair. The pullback on Borel-Moore homology is given by

$$H_*(X) = H_*^{\text{ord}}(\overline{X}, \overline{X} \setminus X) \rightarrow H_*^{\text{ord}}(\overline{X}, \overline{X} \setminus U) = H_*(U).$$

3. Since $i : F \hookrightarrow X$ is a closed embedding, it is proper, and the pushforward i_* on Borel-Moore homology is well-defined. Similarly, $j : U \hookrightarrow X$ is an open embedding and the pullback j^* is well-defined.

Let $X \hookrightarrow M$ be a closed embedding into a smooth manifold M of real dimension m , satisfying the conditions of definition (3) of Borel-Moore homology. Then by Poincaré duality,

$$H_i(X) = H^{m-i}(M, M \setminus X) \quad H_i(F) = H^{m-i}(M, M \setminus F).$$

The set U is open in X and hence locally closed in M . Then there exists a subset $M' \subset M$ such that U is closed in M' . Then the closure of $M \setminus M'$ is contained in the interior of $M \setminus U$. Applying excision to the triple $M \setminus M' \subset M \setminus U \subset M$, the inclusion $\iota : (M', M' \setminus U) \hookrightarrow (M, M \setminus U)$ induces an isomorphism

$$\iota^* : H^k(M, M \setminus U) \xrightarrow{\cong} H^k(M', M' \setminus U)$$

and thus $H^{m-i}(M, M \setminus U) \cong H_i(U)$ by Poincaré duality.

Then the long exact sequence on relative cohomology given by the diagram $F \xleftarrow{i} X \xleftarrow{j} U$

$$\dots \rightarrow H^{k-1}(M, M \setminus U) \rightarrow H^k(M, M \setminus F) \rightarrow H^k(M, M \setminus X) \rightarrow H^k(M, M \setminus U)$$

gives the long exact sequence on Borel-Moore homology via Poincaré duality.

4. Suppose X is irreducible. Let X^{reg} be the Zariski open dense subset of X consisting of the nonsingular points of X . Then X^{reg} is a smooth complex manifold and hence has a canonical orientation, which gives rise to a fundamental class $[X^{\text{reg}}] \in H_m(X^{\text{reg}})$, where m is the real dimension of X .

Since $\dim_{\mathbb{R}}(X \setminus X^{\text{reg}}) \leq m$, then for any $k > m-2$, $H_k(X \setminus X^{\text{reg}}) = 0$ (using some compactification of X and definition (1) of Borel-Moore homology). By the long exact sequence on Borel-Moore homology, this gives an isomorphism $H_m(X) \xrightarrow{\cong} H_m(X^{\text{reg}})$. Define the fundamental class of X , $[X] \in H_m(X)$ to be the inverse image of the fundamental class $[X^{\text{reg}}]$.

If X is reducible, apply the construction above to each irreducible component of X and take the sum of the fundamental classes obtained.

□

Proposition ([CG97], 2.6.14). *Let X be a complex algebraic variety of complex dimension m , and let $X_1, \dots, X_n$ be the irreducible components of X . Then the fundamental classes $[X_1], \dots, [X_n]$ form a basis for the vector space $H_{\text{top}}(X) = H_{2n}(X)$.*

2.7.2 Equivariant derived categories

Let X be an algebraic variety over $\mathbb{C}$. Let G be a linear algebraic group acting on X via the action morphism $a : G \times X \rightarrow X$. Then the G -equivariant derived category of $\mathbb{C}$ -sheaves on X , denoted $D_G(X)$, is the triangulated category equipped with a t -structure and a forgetful functor $\mathcal{F}^G : D_G(X) \rightarrow D^b(X)$ that solves the universal problem given by the following properties:

1. The forgetful functor $\mathcal{F}^G$ preserves t -structures and identifies the heart of $D_G(X)$ with G -equivariant constructible sheaves on X .
2. For a G -equivariant morphism of varieties $f : X \rightarrow Y$, the induced functors f_* , f^* , $f_!$ and $f^!$ lift to functors between $D_G(X)$ and $D_G(Y)$. Similarly, the duality functor D_X lifts to $D_G(X)$, and all the lifted functors satisfy the usual properties.
3. If $A \subset G$ is a normal subgroup of G acting freely on X , then $D_G(X) = D_{G/A}(A \backslash X)$, where $A \backslash X$ denotes the space of A -orbits of X , which is a variety since the A -action is free.

Furthermore, for any closed subgroup $B \subset G$, there exists a canonical forgetful functor $\mathcal{F}_G^B : D_G(X) \rightarrow D_B(X)$.

2.7.3 Equivariant cohomology and Borel-Moore homology

The following definitions are found in [EM97], section 1. Let G be a linear algebraic group acting on an algebraic variety X over $\mathbb{C}$, and let $\pi : X \rightarrow \{pt\}$ be the natural morphism.

Definition. Let $\mathcal{S}$ be a G -equivariant sheaf in $D_G(X)$.

1. The G -equivariant cohomology of $\mathcal{S}$ is

$$H_G^*(X, \mathcal{S}) = \text{Ext}_{D_G(\{pt\})}^*(\mathbb{C}_{pt}, \pi_* \mathcal{S}).$$

2. The compactly supported G -equivariant cohomology of $\mathcal{S}$ is

$$H_{c,G}^*(X, \mathcal{S}) = \text{Ext}_{D_G(X)}^*(\mathbb{C}_{pt}, \pi_! \mathcal{S}).$$

Definition. Let $\mathcal{L}$ be a G -equivariant locally constant sheaf of $\mathbb{C}$ -vector spaces on an algebraic variety X over $\mathbb{C}$, such that the connected components X_i of X are equidimensional. The G -equivariant Borel-Moore homology of $\mathcal{L}$ is given for each $p \in \mathbb{Z}$ by:

$$H_p^{G,\text{BM}}(X, \mathcal{L}) = \bigoplus_i H_G^{p-2\dim X_i}(X, \mathcal{L} \otimes \omega_{X_i})$$

where for a complex algebraic variety Y , $\omega_Y = (Y \rightarrow \{pt\})^! \mathbb{C}_{pt}$ is the dualising sheaf on Y .

2.7.4 Equivariant cohomology for an ind-scheme

The objects considered in this essay, namely affine Springer fibers and Hitchin fibers are ind-schemes that are locally of finite type, hence we give a description of equivariant cohomology in this case, as in [Yun14].

Definition. Let G be a linear algebraic group and let $X = \lim_{\rightarrow} X_n$ be an ind-scheme. A G -action on X is a family of G -actions on each of the X_n such that the transition functions are all G -equivariant morphisms.

For what follows, let G be a linear algebraic group.

Let X be an algebraic space over $\mathbb{C}$ that is locally of finite type. Let $FT(X)$ be the category of open subsets $U \subseteq X$ that are of finite type over $\mathbb{C}$, whose morphisms are open embeddings. Let

$$\underline{D}^b(X) = \varprojlim_{U \in FT(X)} D^b(U).$$

Thus, an object in $\underline{D}^b(X)$ is a system of objects $\mathcal{F}_U \in D^b(U)$ for each open subset U in X that is of finite type over $\mathbb{C}$, together with transition morphisms $\phi_V^U : j^* \mathcal{F}_U \xrightarrow{\sim} \mathcal{F}_V$ for every open embedding $j : V \hookrightarrow U$ satisfying the usual transitivity conditions.

A morphism $\alpha : \mathcal{F} \rightarrow \mathcal{G}$ for objects $\mathcal{F}, \mathcal{G} \in \underline{D}^b(X)$ is a system of maps $\alpha_U : \mathcal{F}_U \rightarrow \mathcal{G}_U$ such that $\alpha_U|_V = \alpha_V$ for all U and V as above.

Let $f : X \rightarrow Y$ be a morphism that is locally of finite type, and let $\mathcal{F} \in \underline{D}^b(X)$, $\mathcal{G} \in \underline{D}^b(Y)$. Then there exist functors:

1. $f^* : \underline{D}^b(Y) \rightarrow \underline{D}^b(X)$ defined as follows: for $U \in FT(X)$, there exists some $V \in FT(Y)$ such that $f(U) \subseteq V$. Denote the restriction of f to U by $f_{U,V} : U \rightarrow V$. Then define

$$(f^* \mathcal{G})_U = f_{U,V}^* \mathcal{G}_V.$$

2. Similarly, with the same notation as above, define the functor $f^! : \underline{D}^b(Y) \rightarrow \underline{D}^b(X)$ by

$$(f^! \mathcal{G})_U = f_{U,V}^! \mathcal{G}_V.$$

3. Suppose f is of finite type. Then define the functor $f_! : \underline{D}^b(X) \rightarrow \underline{D}^b(Y)$ as follows: for $V \in FT(Y)$, $f^{-1}(V) \in FT(X)$. Denote the restriction of f to $f^{-1}(V)$ by $f_V : f^{-1}(V) \rightarrow V$. Then define

$$(f_! \mathcal{F})_V = f_{V,!} \mathcal{F}_{f^{-1}(V)}.$$

However, if f is only locally of finite type, the pushforward $f_!$ will instead be of the form

$$f_! : \underline{D}^b(X) \rightarrow \text{ind} \underline{D}^b(Y)$$

where $\text{ind} \underline{D}^b(Y)$ denotes the category of ind-objects in $\underline{D}^b(Y)$. Then define

$$f_! \mathcal{F} = \varinjlim_{U \in FT(X)} f_{U,!} \mathcal{F}_U$$

where $f_U : U \rightarrow Y$, the restriction of f to U , is of finite type and $f_{U,!}$ is defined as above.

4. Similarly, if f is of finite type, define the pushforward $f_* : \underline{D}^b(X) \rightarrow \underline{D}^b(Y)$ by

$$(f_* \mathcal{F})_V = f_{V,*} \mathcal{F}_{f^{-1}(V)}$$

with analogous notation to previous cases. For f only locally of finite type, we have $f_* : \underline{D}^b(X) \rightarrow \text{pro} \underline{D}^b(Y)$, where $\text{pro} \underline{D}^b(Y)$ denotes the category of pro-objects of $\underline{D}^b(Y)$. Then define

$$f_* \mathcal{F} = \varprojlim_{U \in Ft(X)} f_{U,*} \mathcal{F}_U.$$

In particular for the case here, $Y = \operatorname{Spec} \mathbb{C}$, and X is an ind-scheme, so that $FT(X)$ is indexed by the positive integers. Then the (co)homology of the ind-scheme X is given by:

$$\begin{aligned} H_c^* &= f_! \mathbb{Q}_X H_*(X) = f_! D_X \in \operatorname{ind} D^b(\mathbb{Q}\text{-vector spaces}) \\ H^* &= f_* \mathbb{Q}_X H_*^{\operatorname{BM}}(X) = f_* D_X \in \operatorname{pro} D^b(\mathbb{Q}\text{-vector spaces}) \end{aligned}$$

Supposing now that the ind-scheme $X = \lim_{\rightarrow} X_n$ is equipped with a G -action, the G -equivariant cohomology with compact support of X , where the transition morphisms between the X_n are open embeddings is given by:

$$H_{G,c}^*(X, \mathbb{Q}) = \lim_{\rightarrow} H_{G,c}^*(X_n, \mathbb{Q}).$$

For a strict ind-scheme X (namely, an ind-scheme such that the transition functions are closed embeddings), the G -equivariant cohomology of X is:

$$H_G^*(X, \mathbb{Q}) = \lim_{\leftarrow} H_G^*(X_n, \mathbb{Q}).$$

2.8 Line bundles on the affine flag variety

An alternative way of viewing the affine flag variety $\operatorname{Fl} = LG/I$ is as $\operatorname{Fl} = G_{\operatorname{KM}}/I_{\operatorname{KM}}$, where $I_{\operatorname{KM}} = \mathbb{G}_m^{\operatorname{cen}} \times I \rtimes \mathbb{G}_m^{\operatorname{rot}}$. Then there is a surjection $I_{\operatorname{KM}} \rightarrow A_{\operatorname{KM}} = \mathbb{G}_m^{\operatorname{cen}} \times A \times \mathbb{G}_m^{\operatorname{rot}}$. There is an action of G_{KM} on Fl by left multiplication.

Let $\xi \in \mathbb{X}^*(A_{\operatorname{KM}})$ as a morphism $I_{\operatorname{KM}} \rightarrow A_{\operatorname{KM}} \xrightarrow{\xi} \mathbb{G}_m$, we construct a G_{KM} -equivariant line bundle $\mathcal{L}(\xi) = G_{\operatorname{KM}} \times^{I_{\operatorname{KM}}, \xi} \mathbb{C}_{\xi}$ on Fl , where $\mathbb{C}_{\xi}$ is the one-dimensional $\mathbb{C}$ -vector space acted on by I_{KM} via the character ξ .

Recall in the construction of the determinant line bundle $\det_G$ over the affine flag variety Fl_G used in defining the Kac-Moody group associated to G . This construction can be extended to all (standard) parahoric subgroups P as follows: for P a parahoric subgroup, the determinant line bundle over Fl_P is the line bundle $\det_P$ whose fiber at $gP \in \operatorname{Fl}_P$ is the relative determinant $\det(\operatorname{Ad}(g)\operatorname{Lie}P : \operatorname{Lie}P)$.

There is an action of $P \rtimes \mathbb{G}_m^{\operatorname{rot}}$ on Fl_P given by left multiplication by elements in P and rotation of loops by $\mathbb{G}_m^{\operatorname{rot}}$. Furthermore, for any $p \in P \rtimes \mathbb{G}_m^{\operatorname{rot}}$, we have

$$\det(\operatorname{Ad}(hg)\operatorname{Lie}P : \operatorname{Ad}(h)\operatorname{Lie}P) = (\operatorname{Ad}(hg)\operatorname{Lie}P : \operatorname{Lie}P)$$

since $\operatorname{Ad}(h)\operatorname{Lie}P = \operatorname{Lie}P$. Thus, the determinant line bundle $\det_P$ is equipped with a $P \rtimes \mathbb{G}_m^{\operatorname{rot}}$ -equivariant structure, such that the fiber of $\det_P$ at $h \cdot gP$ is $\det(\operatorname{Ad}(hg)\operatorname{Lie}P : \operatorname{Lie}P)$.

Let $\pi_P : \operatorname{Fl} \rightarrow \operatorname{Fl}_P$ be the projection. Then, for any standard parahoric P (namely, P contains the fixed Iwahori subgroup I), the line bundle $\pi_P^* \det_P$ on Fl is equipped with an $I \rtimes \mathbb{G}_m^{\operatorname{rot}}$ -equivariant structure.

By the construction above, there is an isomorphism of $G(\mathcal{O}_F) \rtimes \mathbb{G}_m^{\operatorname{rot}}$ -equivariant line bundles on Fl

$$\pi_{G(\mathcal{O}_F)}^* \det_G \cong \mathcal{L}(\Lambda_{\operatorname{can}})$$

Let L_P be the Levi factor of a parahoric subgroup P and let $2\rho_P \in \mathfrak{a}_{\operatorname{KM}}^*$ be the sum of the positive roots of L_P . Then the following identity of $I \rtimes \mathbb{G}_m^{\operatorname{rot}}$ -equivariant line bundles on Fl holds [Yun14]:

$$\pi_G^* \det_G = \pi_P^* \det_P \otimes \mathcal{L}(-2\rho_G + 2\rho_P) \otimes \mathcal{O}_{I \rtimes \mathbb{G}_m^{\operatorname{rot}}}(-2\rho_P + 2\rho_{G(\mathcal{O}_F)})$$

where $\mathcal{O}_{I \rtimes \mathbb{G}_m^{\text{rot}}}(\xi)$ is the trivial line bundle on which $I \rtimes \mathbb{G}_m^{\text{rot}}$ acts via the character ξ .

For each affine simple root α_i for $i = 0, \dots, r$, there is a morphism $SL_2 \rightarrow LG$, which induces an embedding $\mathbb{P}^1 \hookrightarrow \text{Fl}$. The cycle class of $\mathbb{P}^1$ gives an element $C_i \in H_2(\text{Fl}, \mathbb{Z})$.

Proposition ([Yun14], Proposition 5.1.5). *For any $\xi \in \mathbb{X}^*(A_{KM})$, and $i = 0, \dots, r$, the following identity holds:*

$$\langle \xi, \alpha_i^\vee \rangle = -\deg(\mathcal{L}(\xi)|C_i). \quad (2.3)$$

Chapter 3

Constructing geometric representations of graded Cherednik algebras

3.1 An $\mathfrak{H}_\nu^{\text{gr}}$ -action on the cohomology of homogeneous affine Springer fibers

The aim is to explain Yun and Oblomkov's construction in [OY14] of an action of $\mathfrak{H}_\nu^{\text{gr}}$ on the $\mathbb{G}_m(\nu)$ -equivariant cohomology of homogeneous Springer fibers.

Let $\nu = \frac{d}{m} > 0 \in \mathbb{Q}$. Recall that for the slope ν , we have a family of homogeneous affine Springer fibers

$$q_\nu : \text{Sp}_\nu \rightarrow \mathfrak{c}(F)_\nu^{\text{rs}}.$$

We construct an action of $\mathfrak{H}_\nu^{\text{gr}}$ on the ind-complex $q_{\nu,!}(\mathbb{Q})$ with cohomology $H_{\mathbb{G}_m(\nu),c}^*(\text{Sp}_\nu, \mathbb{Q})$, by defining the actions of $\mathfrak{a}_{\text{KM}}^*$, s_i and Ω_I separately, then showing that the relations in $\mathfrak{H}_\nu^{\text{gr}}$ hold under these constructions.

Construction 1 (The action of u). Let ϵ be the image of $\delta = -u/\nu$ under the quotient $\mathfrak{H}^{\text{gr}} \rightarrow \mathfrak{H}_\nu^{\text{gr}} = \mathfrak{H}^{\text{gr}}/(u + \nu\delta)$. Note that the $\mathbb{G}_m(\nu)$ -equivariant cohomology of the family of Springer fibers $H_{\mathbb{G}_m(\nu),c}^*(\text{Sp}_\nu, \mathbb{Q})$ is a module over $H_{\mathbb{G}_m(\nu)}^*(\{\text{pt}\}, \mathbb{Q}) \cong \mathbb{Q}[X]$, with $\deg(X) = 2$. Thus, define the action of ϵ as m times the action of the generator $X \in H_{\mathbb{G}_m(\nu)}^2(\{\text{pt}\}, \mathbb{Q})$.

Construction 2 (The action of $\mathfrak{a}_{\text{KM}}^*$). Recall that to a character $\lambda \in \mathbb{X}^*(A_{\text{KM}})$, we associate an A_{KM} -equivariant line bundle $\mathcal{L}_\lambda = G_{\text{KM}} \times^{I_{\text{KM}}, \lambda} \mathbb{C}$ over Fl .

We first consider the finite type case. Let G be a reductive group, choose a maximal torus T and a Borel subgroup B containing it: $T \xrightarrow[\pi]{i} B \hookrightarrow G$. Then G/B is a projective variety: the flag variety Fl .

Given any character $\lambda \in \mathbb{X}^*(T)$, we have a G -equivariant line bundle $\mathcal{L}_\lambda = G \times^B \mathbb{C}_\lambda \in \text{Pic}_{A_{\text{KM}}}(\text{Fl})$, where $\mathbb{C}_\lambda = \mathbb{C}$ with B action given by: $b \cdot z = \lambda(\pi(b))z$ for all $z \in \mathbb{C}$ and $b \in B$, and $G \times^B \mathbb{C}_\lambda$ is the quotient of $G \times \mathbb{C}_\lambda$ by the (free right) action of B defined by: $(g, z) \cdot b = (gb, \lambda(\pi(b))^{-1}z)$.

Thus $\mathcal{L}_\lambda$ is a bundle over Fl as follows:

$$\begin{aligned}\mathcal{L}_\lambda &= G \times^B \mathbb{C}_\lambda \rightarrow \text{Fl} \\ (g, z) \bmod B &\mapsto g \cdot B\end{aligned}$$

All fibers are clearly isomorphic to $\mathbb{C}$, so $\mathcal{L}_\lambda$ is a line bundle and is clearly G -equivariant.

Via Borel's construction, define the T -equivariant cohomology of X to be

$$H_T^*(\text{Fl}) := H^*(ET \times^T \text{Fl})$$

where $ET \rightarrow BT$ is a principal T -bundle over the classifying space BT of T , namely $BT \cong ET/T$.

Note that this definition of T -equivariant cohomology is valid over any space X with a topological group action by any topological group T . For example, if $X = \bullet$ is a point:

$$\begin{aligned}H_T^*(\bullet) &= H^*(ET \times^T \bullet) \\ &= H^*(ET/T) \\ &= H^*(BT) \\ &\cong \mathbb{C}[\mathfrak{t}] \\ &\cong \text{Sym}(\mathfrak{t}^*)\end{aligned}$$

where $H_T^i(\bullet) = \mathbb{C}[\mathfrak{t}]_{\frac{i}{2}}$, the homogeneous elements of $\mathbb{C}[\mathfrak{t}]$ of degree $\frac{i}{2}$.

Lemma. *The G -equivariant cohomology of the flag variety is isomorphic to the T -equivariant cohomology of the point, and furthermore:*

$$H_G^i(\text{Fl}) \cong H_T^i(\bullet) \cong \mathbb{C}[\mathfrak{t}]_{\frac{i}{2}}$$

Under these conditions, there is a map $c_1 : \text{Pic}_G(\text{Fl}) \rightarrow H_T^2(\text{Fl})$, called the *first Chern class*, which has the following properties:

1. The first Chern class is functorial in the following sense: if X and Y are T -varieties, $f : X \rightarrow Y$ is a T -equivariant map, and $\mathcal{L} \in \text{Pic}_T(Y)$, then

$$c_1(f^* \mathcal{L}) = f^* c_1(\mathcal{L}).$$

2. For the G -equivariant line bundle $\mathcal{L}_\lambda$ over Fl seen above, $c_1(\mathcal{L}_\lambda) = \lambda \in \mathfrak{t}^*$, for $\mathfrak{t} = \text{Lie}(T)$, where we note that $\lambda \in \mathbb{X}^*(T) = \text{Hom}(T, \mathbb{C}^\times)$ can be viewed as an element of $\mathfrak{t}^* = \text{Hom}(\mathfrak{t}, \mathbb{C})$.

We now consider the affine case: we have $\text{Fl} = LG/I$, where LG is the loop group and I is an Iwahori subgroup determined by a choice of Borel subgroup $B \subset G$. Recall that Fl is an A_{KM} -equivariant ind-scheme, and that we may equivalently define Fl as $\text{Fl} = G_{\text{KM}}/I_{\text{KM}}$, where $I_{\text{KM}} = \mathbb{G}_m^{\text{cen}} \times I \rtimes \mathbb{G}_m^{\text{rot}}$ and G_{KM} is the Kac-Moody group associated to G . We thus have a surjection $I_{\text{KM}} \rightarrow A_{\text{KM}} = \mathbb{G}_m^{\text{cen}} \times A \times \mathbb{G}_m^{\text{rot}}$. For each character $\lambda \in \mathbb{X}^*(A_{\text{KM}})$, define the set

$$\mathcal{L}_\lambda = G_{\text{KM}} \times^{I_{\text{KM}}} \mathbb{C}_\lambda$$

where again the action of I_{KM} on the G_{KM} component is given by right multiplication, and the action of I_{KM} on $\mathbb{C}_\lambda = \mathbb{C}$ is via the composition $I_{\text{KM}} \twoheadrightarrow A_{\text{KM}} \xrightarrow{\lambda} \mathbb{C}^\times$. As in the finite case, we have a map $\mathcal{L}_\lambda \twoheadrightarrow \text{Fl} = G_{\text{KM}}/I_{\text{KM}}$ given by the natural projection $G_{\text{KM}} \rightarrow \text{Fl}$.

Definition. A *line bundle* over an ind-scheme $X = \bigcup_{n \in \mathbb{N}} X_n$ is a collection of line bundles $\mathcal{L}_n \longrightarrow X_n$ compatible with the transition functions $i_n : X_n \rightarrow X_{n+1}$. We can extend this definition to A -equivariant line bundles for a group A in a natural way.

Lemma. *The set $\mathcal{L}_\lambda$ defined above is an A_{KM} -equivariant line bundle over the ind-scheme Fl .*

We therefore define the first Chern class of $\mathcal{L}_\lambda$ as

$$c_1(\mathcal{L}_\lambda) = (c_1(\mathcal{L}_\lambda|_{\text{Fl}_n}))_{n \in \mathbb{N}} \in H_{A_{\text{KM}}}^2(\text{Fl}) := \varprojlim H_{A_{\text{KM}}}^2(\text{Fl}_n)$$

and by functoriality of c_1 , we have

$$\begin{aligned} i_n^* c_1(\mathcal{L}_\lambda|_{\text{Fl}_{n+1}}) &= c_1(i_n^* \mathcal{L}_\lambda|_{\text{Fl}_{n+1}}) \\ &= c_1(\mathcal{L}_\lambda|_{\text{Fl}_n}) \end{aligned}$$

To get the action of $\xi \in \mathbb{X}^*(A_{\text{KM}})$ on $q_{\nu,!}(\mathbb{Q})$, we want to restrict the A_{KM} -action on Fl to a $\mathbb{G}_m(\nu)$ -action.

For $\nu = d/m \in \mathbb{Q}_{>0}$, there is a one-dimensional sub-torus $\mathbb{G}_m(\nu)$ of A_{KM} , defined via the embedding:

$$\begin{aligned} \mathbb{G}_m(\nu) &\rightarrow A_{\text{KM}} \\ s &\mapsto (s^{d\rho^\vee}, s^m, s^{-d}). \end{aligned}$$

where $\rho^\vee = \frac{1}{2} \sum_{\alpha \in \Delta} \alpha^\vee \in \mathfrak{a}$.

Since A_{KM} acts on the loop group LG , we also have an action of $\mathbb{G}_m(\nu)$ on LG , given by:

$$s \cdot g(t) := \text{Ad}(s^{d\rho^\vee})(g(s^m t))$$

Thus, we restrict the action of A_{KM} on Fl to a $\mathbb{G}_m(\nu)$ action as follows: we can embed

$\mathbb{G}_m(\nu) \hookrightarrow A \times \mathbb{G}_m^{\text{rot}} \hookrightarrow A_{\text{KM}}$ via the cocharacter $d\rho^\vee + m\partial \in \mathbb{X}_*(A) \oplus \mathbb{X}_*(\mathbb{G}_m^{\text{rot}})$.

Therefore for each $\lambda \in \mathbb{X}^*(A_{\text{KM}})$, $\mathcal{L}_\lambda$ has a canonical $\mathbb{G}_m(\nu)$ -equivariant structure and hence we may consider the $\mathbb{G}_m(\nu)$ -equivariant Chern class of $\mathcal{L}_\lambda$.

Finally, define the action of $\lambda \in \mathbb{X}^*(A_{\text{KM}})$ on $q_{\nu,!}(\mathbb{Q})$ by taking the cup product with the $\mathbb{G}_m(\nu)$ -equivariant Chern class:

$$c_1^{\mathbb{G}_m(\nu)}(\mathcal{L}_\lambda) : q_{\nu,!}(\mathbb{Q}) \rightarrow q_{\nu,!}(\mathbb{Q})[2].$$

One possible way of writing the extended affine Weyl group $\widetilde{W}$ is $\widetilde{W} = W_{\text{aff}} \rtimes \Omega_I$ where W_{aff} is generated by all simple reflections corresponding to the affine simple roots (the elements of Δ_{aff} and Ω_I is the stabiliser of the standard alcove associated to the choice of Iwahori subgroup I . The following two constructions define the action of these two subgroups of $\widetilde{W}$.

Construction 3 (The action of simple reflections). Let s_i be a simple reflection in W_{aff} . Let $P \supset I$ be a standard parahoric subgroup. We can decompose P into a semi-direct product $P = U_P \rtimes L_P$, where U_P is the unipotent radical of P , and L_P is the Levi factor of P . In this case, L_P is finite-dimensional, and we consider its associated Springer fibers. Let $\pi_{L_P} : P \rightarrow L_P$ be the natural projection, and let $\pi_{\mathfrak{l}_P} : \text{Lie}(P) \rightarrow \mathfrak{l}_P$, where $\mathfrak{l}_P = \text{Lie}(L_P)$, be its differential. Recall, for $\gamma \in \mathfrak{g}(F)$ regular semisimple, and P a standard parahoric subgroup, the affine Springer fiber $\text{Sp}_{P,\gamma}$ of γ of type P is the following reduced subscheme locally of finite type:

$$\text{Sp}_{P,\gamma} = \{gP \mid g^{-1}\gamma g \in \text{Lie}(P), g \in LG\} \subseteq \text{Fl}_P := LG/P$$

Then we have a map

$$\begin{aligned} \mathrm{ev}_P : \mathrm{Sp}_{P,\gamma} &\rightarrow [\mathfrak{l}_P/L_P] \\ gP &\mapsto \pi_{\mathfrak{l}_P}(g^{-1}\gamma g) \bmod L_P \end{aligned}$$

The parahoric subgroup P is standard, hence contains the Iwahori subgroup I . Let $B_P^I = \pi_{L_P}(I)$ and $\mathfrak{b}_P^I = \pi_{\mathfrak{l}_P}(\mathrm{Lie}(I))$. Then B_P^I is a Borel subgroup of L_P and $\mathfrak{b}_P^I$ is a Borel subalgebra of $\mathfrak{l}_P$. There is an analogous evaluation map on Sp_γ defined by:

$$\begin{aligned} \mathrm{ev}_P^I : \mathrm{Sp}_\gamma &\rightarrow [\mathfrak{b}_P^I/B_P^I] \\ gI &\mapsto \pi_{\mathfrak{l}_P}(g^{-1}\gamma g) \bmod B_P^I \end{aligned}$$

We denote the set of elements of homogeneous elements of slope ν in $\mathfrak{c}^{\mathrm{rs}}(\overline{F})$ by $\mathfrak{c}(F)_\nu^{\mathrm{rs}}$. We can define a family of affine Springer fibers over $\mathfrak{c}(F)_\nu^{\mathrm{rs}}$, $q_{P,\nu} : \mathrm{Sp}_{P,\nu} \rightarrow \mathfrak{c}(F)_\nu^{\mathrm{rs}}$ such that for each $a \in \mathfrak{c}(F)_\nu^{\mathrm{rs}}$, the fiber $q_{P,\nu}^{-1}(a)$ is the Springer fiber $\mathrm{Sp}_{P,\gamma}$ at $\gamma = \kappa(a)$, for κ the Kostant section, denoted by $\mathrm{Sp}_{P,a}$.

For each $\gamma \in \mathfrak{g}(F)$ that is regular semisimple, we have a morphism of ind-schemes:

$$\begin{aligned} \pi_{P,\gamma} : \mathrm{Sp}_\gamma &\rightarrow \mathrm{Sp}_{P,\gamma} \\ gI &\mapsto gP \end{aligned}$$

that is well-defined since $I \subset P$, hence $g^{-1}\gamma g \in \mathrm{Lie}(I)$ implies $g^{-1}\gamma g \in \mathrm{Lie}(P)$. These maps on fibers induce a map on families of fibers:

$$\pi_{P,\nu} : \mathrm{Sp}_\nu \rightarrow \mathrm{Sp}_{P,\nu}$$

Define a morphism on quotient stacks $p_{\mathfrak{l}_P} : [\mathfrak{b}_P^I/B_P^I] \rightarrow [\mathfrak{l}_P/L_P]$ induced from the inclusion $\mathfrak{b}_P^I \hookrightarrow \mathfrak{l}_P$. We therefore have the following commutative diagram:

$$\begin{array}{ccc} \mathrm{Sp}_\nu & \xrightarrow{\mathrm{ev}_P^I} & [\mathfrak{b}_P^I/B_P^I] \\ \downarrow \pi_{P,\nu} & & \downarrow p_{\mathfrak{l}_P} \\ \mathrm{Sp}_{P,\nu} & \xrightarrow{\mathrm{ev}_P} & [\mathfrak{l}_P/L_P] \end{array} \quad (3.1)$$

The morphism $p_{\mathfrak{l}_P}$ in diagram (3.1) is the Grothendieck simultaneous resolution of $\mathfrak{l}_P$ modulo the adjoint action of L_P .

Lemma. *By induction, there is an isomorphism of stacks*

$$[\mathfrak{b}_P^I/B_P^I] \stackrel{\mathrm{Ind}}{\cong} [\mathfrak{b}_P^I \times^{B_P^I} L_P/L_P]$$

Let $i : \mathfrak{b}_P^I \hookrightarrow \mathfrak{l}_P$ be the natural inclusion. Define $p'_{\mathfrak{l}_P}$ to be the morphism of quotient stacks induced by

$$\begin{aligned} \mathfrak{b}_P^I \times^{B_P^I} L_P &\rightarrow \mathfrak{l}_P \\ (x, g) &\mapsto \mathrm{ad}(g^{-1})(i(x)) \end{aligned}$$

The following diagram commutes (concretely, compose $p_{\mathfrak{l}_P}$ with the isomorphism from the lemma above):

$$\begin{array}{ccc} [\mathfrak{b}_P^I/B_P^I] & \xrightarrow{\cong} & [\mathfrak{b}_P^I \times^{B_P^I} L_P/L_P] \\ \downarrow p_{\mathfrak{l}_P} & \swarrow p'_{\mathfrak{l}_P} & \\ [\mathfrak{l}_P/L_P] & & \end{array}$$

Denote $\tilde{\mathfrak{b}}_P^I := \mathfrak{b}_P^I \times^{B_P^I} L_P$. Diagram (3.1) becomes

$$\begin{array}{ccc} \text{Sp}_\nu & \xrightarrow{\text{ev}_P^I} & [\tilde{\mathfrak{b}}_P^I/L_P] \\ \downarrow \pi_{P,\nu} & & \downarrow p'_{\mathfrak{l}_P} \\ \text{Sp}_{P,\nu} & \xrightarrow{\text{ev}_P} & [\mathfrak{l}_P/L_P] \end{array} \quad (3.2)$$

Denote the constant sheaf over a space X with stalks equal to the field $\mathbb{Q}$ by $\mathbb{Q}_X$. Noting that under the induction functor

$$\begin{aligned} \text{Ind} : D_{\mathbb{G}_m(\nu)}^b(\mathfrak{b}_P^I) &\rightarrow D_{\mathbb{G}_m(\nu)}^b(\tilde{\mathfrak{b}}_P^I) \\ \mathbb{Q}_{\mathfrak{b}_P^I} &\mapsto \mathbb{Q}_{\tilde{\mathfrak{b}}_P^I}, \end{aligned}$$

and by the definition of the morphism $p'_{\mathfrak{l}_P}$, we have $p_{\mathfrak{l}_P,*}(\mathbb{Q}_{\mathfrak{b}_P^I}) = p'_{\mathfrak{l}_P,*}(\mathbb{Q}_{\tilde{\mathfrak{b}}_P^I})$.

The aim now is to reduce to the finite case, in particular the case where G is of semisimple rank 1. To this end, let P_i be the standard parahoric subgroup whose Levi factor L_i has root system $\{\pm\alpha_i\}$. Equivalently, this means that $P_i \supset I$ is defined by $\text{Lie}(P_i) = \text{Lie}(I) \oplus (\mathfrak{g}(F))_{-\alpha_i}$, where α_i is an affine simple root and $\mathfrak{g}(F)_{-\alpha_i}$ denotes the $-\alpha_i$ root space. We thus have the corresponding Weyl group $W_{L_i} = \langle s_i \rangle$, where s_i is the simple reflection corresponding to the root α_i . Denote $\text{Lie}(L_i)$ by $\mathfrak{l}_i$.

Denote $\tilde{\mathfrak{b}}_i := \tilde{\mathfrak{b}}_{P_i}^I$ and $Q_i := [\mathfrak{b}_{P_i}^I/B_{P_i}^I]$.

Applying proper base change to (3.1) for $P = P_i$, we obtain the following isomorphism in the equivariant derived category $D_{\mathbb{G}_m(\nu)}^b(\text{Sp}_{P_i,\nu})$:

$$\pi_{P_i,\nu,*}(\mathbb{Q}_{\text{Sp}_\nu}) \cong \text{ev}_P^* p_{\mathfrak{l}_i,*}(\mathbb{Q}_{Q_i}). \quad (3.3)$$

Furthermore, using the discussion above and applying base change, this isomorphism becomes:

$$\pi_{P_i,\nu,*}(\mathbb{Q}_{\text{Sp}_\nu}) \cong \text{ev}_{L_i}^* p'_{\mathfrak{l}_i,*}(\mathbb{Q}_{\tilde{\mathfrak{b}}_i}). \quad (3.4)$$

The following is a well-known result from Springer theory:

Proposition. *The subgroup of the affine Weyl group W_{aff} generated by the involution s_i acts on the complex $p_{\mathfrak{l}_i,*}(\mathbb{Q}_{Q_i})$ and hence on $p'_{\mathfrak{l}_i,*}(\mathbb{Q}_{\tilde{\mathfrak{b}}_i})$ in $D_{L_i}^b(\mathfrak{l}_i)$.*

Proof. Let $\tilde{\mathfrak{l}}_i = L_i \times^{B_i} \mathfrak{b}_i$. This is a vector bundle $\tilde{\mathfrak{l}}_i \rightarrow L_i/B_i \cong \mathbb{P}^1$. The Steinberg variety is

$$\begin{aligned} \text{St}_i &= \tilde{\mathfrak{l}}_i \times_{\mathfrak{l}_i} \tilde{\mathfrak{l}}_i \\ &= \{(g \cdot B_i, y; g' \cdot B_i, y') \mid \text{ad}(g^{-1})(y), \text{ad}(g'^{-1})(y') \in \mathfrak{b}_i \text{ and } y = y'\} \\ &\cong \{(g \cdot B_i, g' \cdot B_i, y) \in L_i/B_i \times L_i/B_i \times \mathfrak{l} \mid \text{ad}(g^{-1})(y), \text{ad}(g'^{-1})(y) \in \mathfrak{b}_i\} \end{aligned}$$

This gives the following Cartesian square:

$$\begin{array}{ccc}
& \text{St}_i & \\
q_1 \swarrow & & \searrow q_2 \\
\tilde{\mathbb{I}}_i & & \tilde{\mathbb{I}}_i \\
p_1 \searrow & & \swarrow p_2 \\
& \mathbb{I}_i &
\end{array} \tag{3.5}$$

where q_1 and q_2 are the natural projections onto the first and second component respectively, and $p_1 = p_2 = q$. The morphisms q, q_1 and q_2 are proper, so the fibers are compact. Furthermore, there is an action of L_i on each of the varieties in the above Cartesian square that commutes with each of the morphisms.

The method to describe the action of s_i is as follows: given a closed $\tilde{L}_i = L_i \times \mathbb{C}^\times$ -invariant subvariety Y of the Steinberg variety St_i , consider its fundamental class $[Y]$ in the $\tilde{L}_i$ -equivariant Borel-Moore homology of the Steinberg variety $H_{*, \tilde{L}_i}^{\text{BM}}(\text{St}_i, \mathbb{Q})$. To $[Y]$ we associate an element of $\text{Hom}_{D_{\tilde{L}_i}^b(\text{St}_i)}(q_1^* \mathbb{Q}_{\tilde{\mathbb{I}}_i}, q_2^! \mathbb{Q}_{\tilde{\mathbb{I}}_i})$, and from this an element of $\text{End}(q_* \mathbb{Q}_{\mathbb{I}_i})$. We determine the subvariety Y after the following discussion.

To obtain a map $\beta : \text{Hom}_{D_{\tilde{L}_i}^b(\text{St}_i)}(q_1^* \mathbb{Q}_{\tilde{\mathbb{I}}_i}, q_2^! \mathbb{Q}_{\tilde{\mathbb{I}}_i}) \xrightarrow{\sim} \text{End}(q_* \mathbb{Q}_{\mathbb{I}_i})$, note that by the adjunction between the pushforward and pullback:

$$\text{Hom}_{D_{\tilde{L}_i}^b(\text{St}_i)}(q_1^* \mathbb{Q}_{\tilde{\mathbb{I}}_i}, q_2^! \mathbb{Q}_{\tilde{\mathbb{I}}_i}) = \text{Hom}_{D_{\tilde{L}_i}^b(\text{St})}(\mathbb{Q}_{\tilde{\mathbb{I}}_i}, q_{1,*} q_2^! \mathbb{Q}_{\tilde{\mathbb{I}}_i})$$

Furthermore, by base change in the Cartesian square above, we have $q_{1,*} q_2^! \mathbb{Q}_{\tilde{\mathbb{I}}_i} = p_1^! p_{2,*} \mathbb{Q}_{\tilde{\mathbb{I}}_i}$. Thus:

$$\begin{aligned}
\text{Hom}_{D_{\tilde{L}_i}^b(\text{St}_i)}(q_1^* \mathbb{Q}_{\tilde{\mathbb{I}}_i}, q_2^! \mathbb{Q}_{\tilde{\mathbb{I}}_i}) &= \text{Hom}_{D_{\tilde{L}_i}^b(\text{St}_i)}(\mathbb{Q}_{\tilde{\mathbb{I}}_i}, p_1^! p_{2,*} \mathbb{Q}_{\tilde{\mathbb{I}}_i}) \\
&= \text{Hom}_{D_{\tilde{L}_i}^b(\mathbb{I}_i)}(p_{1,!} \mathbb{Q}_{\tilde{\mathbb{I}}_i}, p_{2,*} \mathbb{Q}_{\tilde{\mathbb{I}}_i}) \\
&= \text{Hom}_{D_{\tilde{L}_i}^b(\mathbb{I}_i)}(q_* \mathbb{Q}_{\tilde{\mathbb{I}}_i}, q_* \mathbb{Q}_{\tilde{\mathbb{I}}_i})
\end{aligned}$$

where the last equality is due to $p_{1,!} = p_{1,*}$ since p_1 is proper, and $p_1 = p_2 = q$.

From [Lus95] p225:

$$\text{Hom}_{D_{\tilde{L}_i}^b(\text{St}_i)}(q_1^* \mathbb{Q}_{\tilde{\mathbb{I}}_i}, q_2^! \mathbb{Q}_{\tilde{\mathbb{I}}_i}) = H_{\tilde{L}_i}^0(\text{St}_i, D(D(q_2^! \mathbb{Q}_{\tilde{\mathbb{I}}_i}) \otimes q_1^* \mathbb{Q}_{\tilde{\mathbb{I}}_i}))$$

Then $D(q_2^! \mathbb{Q}_{\tilde{\mathbb{I}}_i}) = q_2^* D \mathbb{Q}_{\tilde{\mathbb{I}}_i}$, $q_1^* \mathbb{Q}_{\tilde{\mathbb{I}}_i} = \mathbb{Q}_{\text{St}_i}$, and the constant sheaf is the unit for the tensor product on sheaves. Thus $D q_2^* D \mathbb{Q}_{\tilde{\mathbb{I}}_i} = q_2^! D^2 \mathbb{Q}_{\tilde{\mathbb{I}}_i} = q_2^! \mathbb{Q}_{\tilde{\mathbb{I}}_i}$. Therefore,

$$\text{Hom}_{D_{\tilde{L}_i}^b(\text{St}_i)}(q_1^* \mathbb{Q}_{\tilde{\mathbb{I}}_i}, q_2^! \mathbb{Q}_{\tilde{\mathbb{I}}_i}) = H_{\tilde{L}_i}^0(\text{St}_i, q_2^! \mathbb{Q}_{\tilde{\mathbb{I}}_i})$$

By [CG97] p. 427, $q_2^! \mathbb{Q}_{\tilde{\mathbb{I}}_i} = D \text{St}_i$, where $D \text{St}_i = D(\mathbb{Q}_{\text{St}_i})$ is the dualising complex for the Steinberg variety.

Conclusion:

$$\text{Hom}_{D_{\tilde{L}_i}^b(\text{St}_i)}(q_1^* \mathbb{Q}_{\tilde{\mathbb{I}}_i}, q_2^! \mathbb{Q}_{\tilde{\mathbb{I}}_i}[-d]) = H_{\tilde{L}_i}^{-d}(\text{St}_i, D \text{St}_i)$$

since for a scheme X , with a group scheme G acting on X , $H_i^{G, \text{BM}}(X, \mathbb{Q}) = H_G^{-i}(X, D_X)$, $H_{\tilde{L}_i}^{-d}(\text{St}_i, D \text{St}_i) = H_d^{\tilde{L}_i, \text{BM}}(\text{St}_i, \mathbb{Q})$.

Therefore, to get the s_i action on $q^*\mathbb{Q}$, it remains to choose the appropriate closed subvariety of the Steinberg variety. For this, decompose the Steinberg variety

$$\text{St}_i = \{(g_1 B_i, g_2 B_i, y) | y \in \text{Ad}(g_1)(\mathfrak{b}_i) \cap \text{Ad}(g_2)(\mathfrak{b}_i)\}$$

into its two irreducible components: $\text{St}_i = \text{St}_i^+ \cup \text{St}_i^-$, where St_i^+ is the diagonal component and St_i^- is the non-diagonal component:

$$\begin{aligned} \text{St}_i^+ &= \{(g B_i, g B_i, y) | y \in \text{Ad}(g)(\mathfrak{b}_i)\} \cong \tilde{\mathfrak{l}}_i \\ \text{St}_i^- &= \overline{\{(g B_i, g' B_i, y) | g B \neq g' B \text{ and } y \in \text{Ad}(g)(\mathfrak{b}_i) \cap \text{Ad}(g')(\mathfrak{b}_i)\}} \end{aligned}$$

Let $T \subset B_i \subset L_i$ be a maximal torus. There is a morphism:

$$\begin{aligned} \Psi : H_*^{T \times \mathbb{C}^\times}(\text{St}) &\rightarrow \text{End}(H_*^{T \times \mathbb{C}^\times}(\tilde{\mathfrak{l}})) \\ c &\mapsto q_{1,*}(c \cap q_2^*(\bullet)) \end{aligned}$$

where $\cap$ denotes the intersection product for the smooth variety $\tilde{\mathfrak{l}}_i \times \tilde{\mathfrak{l}}_i \supset \text{St}_i$ and the projections p_1 and p_2 were defined in (3.5).

The Weyl group associated to the Levi subgroup L_i is $W_i = \{1, s_i\}$, where s_i is a reflection. Thus to determine the action of the Weyl group on $H_*^{T \times \mathbb{C}^\times}(\tilde{\mathfrak{l}}_i)$, define the endomorphism S_i on $H_*^{T \times \mathbb{C}^\times}(\tilde{\mathfrak{l}}_i)$ such that under the morphism Ψ above:

$$\Psi([\text{St}_i^-]) = S_i$$

It is clear from the definition of the morphism Ψ that $\Psi([\text{St}_i^+]) = q_{1,*}([\text{St}_i^+ \cap q_2^*(\bullet)]) = \text{id}$.

The convolution product in Borel-Moore homology gives $[\text{St}_i^-] * [\text{St}_i^-] = [\text{St}_i^+]$, and therefore

$$S_i^2 = \Psi([\text{St}_i^-] * [\text{St}_i^-]) = \Psi([\text{St}_i^+]) = \text{id}$$

(further details can be found in [CG97]) hence S_i is a reflection and determines the action of the subgroup generated by the involution s_i on $p'_{L_i,*}(\mathbb{Q}_{\tilde{\mathfrak{b}}_i})$. □

By definition of the families of Springer fibers, we can complete (3.2) to the following commutative diagram:

$$\begin{array}{ccc} \text{Sp}_\nu & \xrightarrow{\text{ev}_{P_i}^I} & [\tilde{\mathfrak{b}}_{P_i}^{B_{P_i}^I} \times L_{P_i}/L_{P_i}] \\ \swarrow q_\nu & \downarrow \pi_{P_i, \nu} & \downarrow p'_{\mathfrak{l}_{P_i}} \\ \mathfrak{c}(F)_\nu^{\text{rs}} & & \\ \swarrow q_{P_i, \nu} & \downarrow & \\ \text{Sp}_{P_i, \nu} & \xrightarrow{\text{ev}_{P_i}} & [l_{P_i}/L_{P_i}] \end{array} \quad (3.6)$$

By (3.6), there is an isomorphism

$$q_{\nu,*}(\mathbb{Q}_{\text{Sp}_\nu}) \cong q_{P_i, \nu,*} \pi_{P_i, \nu,*}(\mathbb{Q}_{\text{Sp}_\nu})$$

By functoriality, the s_i action on $p'_{\mathfrak{l}_i,*}(\mathbb{Q}_{\tilde{\mathfrak{b}}_i})$ induces an s_i action on $q_{\nu,*}(\mathbb{Q}_{\text{Sp}_\nu})$. This holds for all the affine simple reflections s_i , and this hence completes the construction of an action of the affine Weyl group W_{aff} on $q_{\nu,*}$.

Construction 4 (The Ω_I action). First, define an action of $\Omega_I = N_G(I)/I$ on the family of Springer fibers Sp_ν by right multiplication. Let $\gamma = \kappa(a)$ for some $a \in \mathfrak{c}(F)_\nu^{\mathrm{rs}}$. For an element $\omega \in \Omega_I$, let $\dot{\omega} \in N_G(I)$ be a representative of ω , then

$$\begin{aligned} \omega : \mathrm{Sp}_\gamma &\rightarrow \mathrm{Sp}_\gamma \\ gI &\mapsto g\dot{\omega}I \end{aligned}$$

This is a well-defined automorphism of Sp_γ , since for any $gI \in \mathrm{Sp}_\gamma$, we have $\mathrm{Ad}(g^{-1})\gamma \in \mathrm{Lie}(I)$, and since $\dot{\omega} \in N_G(I)$, we again have $\mathrm{Ad}((g\dot{\omega})^{-1})\gamma \in \mathrm{Lie}(I)$. Extending this definition to the family Sp_ν , we get an automorphism on Sp_ν that preserves each fiber and it is clear that this definition is independent of the choice of representative of ω in $N_G(I)$.

Lemma. *This action commutes with the $\mathbb{G}_m(\nu)$ -action on Sp_ν .*

Since the defined action of Ω_I commutes with the $\mathbb{G}_m(\nu)$ -action on Sp_ν , this induces an Ω_I -action on $q_{\nu,!}(\mathbb{Q}_{\mathrm{Sp}_\nu}) \in D_{\mathbb{G}_m(\nu)}^b(\mathfrak{c}(F)_\nu^{\mathrm{rs}})$.

This completes the construction of the $\widetilde{W}$ -action on $q_{\nu,!}(\mathbb{Q}_{\mathrm{Sp}_\nu})$. Together, these constructions give us the following theorem:

Theorem. *Constructions (1) - (4) above define an action of the graded Cherednik algebra $\mathfrak{H}_\nu^{\mathrm{gr}}$ on $q_{\nu,!}(\mathbb{Q}_{\mathrm{Sp}_\nu}) \in D_{\mathbb{G}_m(\nu)}^b(\mathfrak{c}(F)_\nu^{\mathrm{rs}})$. The action of $B_{KM} \in \mathrm{Sym}^2(\mathfrak{a}_{KM}^*)^{\widetilde{W}}$ on $q_{\nu,!}(\mathbb{Q})$ is through multiplication by $\nu^2 \mathbb{B}(\rho^\vee, \rho^\vee) \epsilon^2 \in H_{\mathbb{G}_m(\nu)}^4(pt)$.*

3.2 Proof

3.2.1 Relation (G1)

In this context, relation **G1** is translated to the requirement that the action of $\epsilon = -u/\nu$ commutes with the action of all other elements in $\mathfrak{H}_\nu^{\mathrm{gr}}$. This is guaranteed by the construction of the action of ϵ .

3.2.2 Relation (G2)

First, it is clear by construction that multiplication in $\mathrm{Sym}(\mathfrak{a}_{KM}^*)$ lifts compatibly to the action on $q_{\nu,!}(\mathbb{Q})$.

In [Lus96], Lusztig showed that the restriction of the actions of the individual $W_i = \{1, s_i\}$ are compatible on intersections and thus give an action of W_{aff} on $H^*(\mathrm{Sp}_\gamma)$.

It remains to show that the constructions (3) and (4) of the separate W and Ω_I actions are compatible and hence form an action of the extended affine Weyl group $\widetilde{W}$.

Construction (4) gives a left action for any $w \in \Omega_I$ on $H_*(\mathrm{Sp}_\gamma)$:

$$R_{w,*}^- 1 : H_*(\mathrm{Sp}_\gamma) \rightarrow H_*(\mathrm{Sp}_\gamma).$$

Furthermore, right multiplication by an element $w \in N_G(I)$ sends a standard parahoric subgroup P to a standard parahoric $w^{-1}Pw$, and gives an isomorphism $F_{P,\gamma} \xrightarrow{\cong} F_{w^{-1}Pw,\gamma}$.

Therefore the following diagram commutes:

$$\begin{array}{ccc}
\mathrm{Sp}_\gamma & \xrightarrow{R_w} & \mathrm{Sp}_\gamma \\
\downarrow \nu_P & & \downarrow \nu_{w^{-1}Pw} \\
\mathrm{Sp}_{P,\gamma} & \xrightarrow{R_w} & \mathrm{Sp}_{w^{-1}Pw,\gamma}
\end{array}$$

and $R_{w,*}^{-1}$ intertwines the action of $W_{w^{-1}Pw}$ and W_P on $H_*(\mathrm{Sp}_\gamma)$. Letting $P = P_i$, this gives the compatibility of the W_{aff} and Ω_I actions.

3.2.3 Relation (G3)

In construction (3), the action of simple reflections was given by embedding the Weyl group W_i corresponding to a Levi subgroup L_i into $\mathrm{End}(H_*^{T \times \mathbb{C}^\times}(\tilde{\mathfrak{l}}_i))$, where $T \subset B_i \subset L_i$ are a maximal torus and Borel subgroup respectively and $\tilde{\mathfrak{l}}_i = L_i \times^{B_i} \mathfrak{b}_i$.

Recall that $x_\xi \in H_T^2(\tilde{\mathfrak{l}}_i)$ is the first Chern class of a line bundle on L_i/B_i determined by a character $\xi \in \mathbb{X}^*(T)$.

The aim is to show that:

$$x_\xi s_i - s_i x_{s_i(\xi)} = \langle \xi, \alpha_i^\vee \rangle u$$

where α_i is the simple root in Δ_{aff} corresponding to the reflection s_i and u is the generator of $H_{T \times \mathbb{C}^\times}^*(\tilde{\mathfrak{l}}_i)$.

Under the map Ψ , to show the relation in $\mathrm{End}(H_*^{T \times \mathbb{C}^\times}(\tilde{\mathfrak{l}}_i))$ it suffices to show the following relation in $H_*^{T \times \mathbb{C}^\times}(\mathrm{St}_i)$.

Proposition ([Yun11] 6.4.2). *In $H_*^{T \times \mathbb{C}^\times}(\mathrm{St}_i)$, the following identity holds for all $\xi \in \mathfrak{a}_{KM}^*$ and all affine simple reflections s_i .*

$$(q_1^*(x_\xi) - q_2^*(x_\xi)) \cap [\mathrm{St}_i^-] = \langle \xi, \alpha_i^\vee \rangle u \cap [\mathrm{St}_i^+] \quad (3.7)$$

Proof. First, suppose $\xi \in \mathfrak{a}^*$.

By the decomposition of $\mathbb{X}^*(A \times \mathbb{G}_m^{\mathrm{rot}}) \otimes \mathbb{Q} = (\mathbb{X}^*(A \times \mathbb{G}_m^{\mathrm{rot}})^{s_i} \otimes \mathbb{Q}) \oplus \mathbb{Q}\alpha_i$, there are two cases:

1. $s_i(\xi) = \xi$

In this case, $\langle \xi, \alpha_i^\vee \rangle = 0$ and it remains to show that $(q_1^*(x_\xi) - q_2^*(x_\xi)) \cap [\mathrm{St}_i^-] = 0$.

Since $s_i(\xi) = \xi$, this means that $\xi \in \mathbb{X}^*(T)^{s_i} = \mathbb{X}^*(T)^{W_i} \xleftarrow{\simeq} \mathbb{X}^*(L_i) \xrightarrow{\mathrm{res}} \mathbb{X}^*(T)$, hence the line bundle associated to ξ is trivial:

$$\mathcal{L}(\xi) = L_i \times^{B_i} \mathbb{C}_\xi \xrightarrow{\simeq} L_i/B_i \times \mathbb{C}_\xi$$

Lemma. *Recall in the Cartesian square defining the Steinberg variety (3.5), $p_1 = p_2 = q : \tilde{\mathfrak{l}}_i \rightarrow \mathfrak{l}_i$. Then $x_\xi = q^*(\gamma_\xi)$ for $\gamma_\xi \in H_{L_i}^2(\mathfrak{l}_i)$.*

Proof. By definition, $x_\xi = c_1(\mathcal{N}(\xi))$, where

$$\begin{aligned}
\mathcal{N}(\xi) &= L_i \times^{B_i} (\mathfrak{b}_i \times \mathbb{C}_\xi) \\
&\cong (L_i \times^{B_i} \mathfrak{b}_i) \times \mathbb{C}_\xi \text{ since } \mathcal{L}(\xi) \text{ is trivial} \\
&= \tilde{\mathfrak{l}}_i \times \mathbb{C}_\xi
\end{aligned}$$

Hence $\mathcal{N}(\xi)$ is also a trivial line bundle, and the following is a Cartesian diagram:

$$\begin{array}{ccc} \tilde{\mathfrak{l}}_i & \xrightarrow{p} & L_i/B_i \\ q \downarrow & & \downarrow a \\ \mathfrak{l}_i & \xrightarrow{b} & \{pt\} \end{array}$$

Since $\mathcal{L}(\xi)$ is a trivial line bundle, $\mathcal{L}(\xi) = a^*(\mathbb{C}_{\{pt\}})$ and hence

$$\mathcal{N}(\xi) = p^* a^*(\mathbb{C}_{\{pt\}}) = q^*(b^*(\mathbb{C}_{\{pt\}}))$$

and x_ξ is the pullback of an element in $H_{L_i}^2(\mathfrak{l}_i)$. $\square$

By the lemma,

$$\begin{aligned} q_1^*(x_\xi) - q_2^*(x_\xi) &= q_1^*(q^*(\gamma_\xi)) - q_2^*(q^*(\gamma_\xi)) \\ &= (q \circ q_1)^*(\gamma_\xi) - (q \circ q_2)^*(\gamma_\xi) \\ &= 0 \end{aligned}$$

where the final equality holds by the Cartesian diagram defining the Steinberg variety.

2. $\underline{s_i(\xi) = -\xi}$

In this case, $\xi = \alpha_i$ and $\langle \alpha_i^\vee, \xi \rangle = 2$.

Let $p: \tilde{\mathfrak{l}}_i \rightarrow L_i/B_i \cong \mathbb{P}^1$ be the line bundle given by $\tilde{\mathfrak{l}}_i = L_i \overset{B_i}{\times} \mathfrak{b}_i$ and let $x_{\alpha_i} = c_1(\mathcal{N}(\alpha_i))$ where $\mathcal{N}(\alpha_i) = L_i \overset{B_i}{\times} (\mathfrak{b}_i \times \mathbb{C}_{\alpha_i})$ is a line bundle over $\tilde{\mathfrak{l}}_i$. Let $L_i^{\text{ad}} = L_i/Z(L_i)$ be the adjoint form of L_i .

$\mathcal{N}(\alpha_i)$ is the pullback of the canonical line bundle $\omega_{\mathbb{P}^1}$ on $\mathbb{P}^1$ via the morphism p , where the canonical bundle ω_X of a nonsingular algebraic variety X of dimension n over a field k is the n -th exterior power of the cotangent bundle on X .

Let $h: \text{St}_i^- \rightarrow \mathbb{P}^1 \times \mathbb{P}^1$ be the $L_i \times \mathbb{G}_m$ -equivariant morphism defined via $\text{St}_i^- \xrightarrow{(q_1, q_2)} \tilde{\mathfrak{l}}_i \times \tilde{\mathfrak{l}}_i \rightarrow \mathbb{P}^1 \times \mathbb{P}^1$ by

$$h(g_1 \cdot B_i, g_2 \cdot B_i, y) = (g_1 \cdot B_i, g_2 \cdot B_i) \in L_i/B_i \times L_i/B_i \cong \mathbb{P}^1 \times \mathbb{P}^1.$$

By definition of h , there is a commutative diagram:

$$\begin{array}{ccccccc} \mathbb{P}^1 & \xleftarrow{p} & \tilde{\mathfrak{l}}_i & \xleftarrow{q_1} & \text{St}_i^- & \xrightarrow{q_2} & \tilde{\mathfrak{l}}_i & \xrightarrow{p} & \mathbb{P}^1 \\ & & & & \downarrow h & & & & \\ & & & & \mathbb{P}^1 \times \mathbb{P}^1 & & & & \end{array} \quad (3.8)$$

r_1 r_2

where r_1 and r_2 are the projections onto the first and second component respectively.

Through the isomorphism $T^*(\mathcal{B} \times \mathcal{B}) \cong T^*(\mathcal{B}) \times T^*(\mathcal{B})$, there is a correspondence:

$$\omega_{\mathbb{P}^1 \times \mathbb{P}^1} = r_1^* \omega_{\mathbb{P}^1} - r_2^* \omega_{\mathbb{P}^1}$$

hence

$$h^* c_1(\omega_{\mathbb{P}^1 \times \mathbb{P}^1}) = h^* c_1(r_1^* \omega_{\mathbb{P}^1} - r_2^* \omega_{\mathbb{P}^1})$$

by functoriality of the first Chern class and the commutativity of diagram (3.8) above:

$$h^* c_1(\omega_{\mathbb{P}^1 \times \mathbb{P}^1}) = q_1^* c_1 \mathcal{N}(\alpha_i) - q_2^* c_1 \mathcal{N}(\alpha_i)$$

since $\mathcal{N}(\alpha) = p^*(\omega_{\mathbb{P}^1})$. Hence, equation (3.7) becomes:

$$h^* c_1(\omega_{\mathbb{P}^1 \times \mathbb{P}^1}) \cap [\text{St}_i^-] = 2u \cap [\text{St}_i^+] \quad (3.9)$$

The aim is now to show that both sides of equation (3.9) are equal to a certain cycle class $2[\text{St}^{\text{nil}}] \in H_*^{L \times \mathbb{G}_m}(\text{St})$, where St^{nil} is defined below. For a real oriented vector bundle $p : E \rightarrow B$ of rank r over a manifold B , denote the Euler class of E by $\text{eu}(E) \in H^{2n}(B)$.

We use the following well-known facts without proof:

Lemma. (a) Let $\Delta : \mathbb{P}^1 \rightarrow \mathbb{P}^1 \times \mathbb{P}^1$ be the natural embedding, so that $\Delta(\mathbb{P}^1)$ is the diagonal in $\mathbb{P}^1 \times \mathbb{P}^1$. Then by Poincaré duality

$$c_1(\omega_{\mathbb{P}^1 \times \mathbb{P}^1}) \cap [\mathbb{P}^1 \times \mathbb{P}^1] = 2[\Delta(\mathbb{P}^1)].$$

(b) Let X be smooth, $p : E \rightarrow X$ a vector bundle over X and $Y \subseteq X$ a closed subvariety.

Let $E|_Y = p^{-1}(Y)$. Then for $\omega \in H^*(X)$, $\omega \cap [X] = [Y]$ implies $p^*(\omega) \cap [E] = [E|_Y]$.

(c) The morphism $h : \text{St}_i^- \rightarrow \mathbb{P}^1 \times \mathbb{P}^1$ is a vector bundle.

(d) The restriction of h to $\text{St}_i^-|_{\Delta(\mathbb{P}^1)} \rightarrow \Delta(\mathbb{P}^1)$ is a vector bundle over $\Delta(\mathbb{P}^1) \cong \mathbb{P}^1$.

(e) As vector bundles over $\mathbb{P}^1 \cong L_i/B_i$, $\text{St}_i^+ \cong L_i \times^{B_i} \mathfrak{b}_i$.

(f) Let X be smooth and let $p : E \rightarrow X$ be a complex vector bundle over X , with $\text{rk} E = n$. Let $s : X \rightarrow E$ be a section of the vector bundle and let $Y = s^{-1}(0) \subseteq X$. If Y is smooth of codimension n then $\text{eu}(E) \cap [X] = [Y]$.

Furthermore, the results above hold in the equivariant case, when applicable.

Define $\text{St}_i^{\text{nil}} := h^{-1}(\Delta(\mathbb{P}^1)) = \text{St}_i^-|_{\Delta(\mathbb{P}^1)}$. Then by part (d) of the lemma, St_i^{nil} is a vector bundle over $\mathbb{P}^1 \cong L_i/B_i$.

Part (c) of the lemma says that the morphism h is smooth, and hence we can take the pull back of the equation in part (a) of the lemma along h in Borel-Moore homology to get:

$$2[\text{St}_i^{\text{nil}}] = h^* c_1(\omega_{\mathbb{P}^1 \times \mathbb{P}^1}) \cap [\text{St}_i^-].$$

For the other side of the equation, let $T^{\text{ad}} \subset B_i^{\text{ad}} \subset L_i^{\text{ad}}$ be a maximal torus and Borel subgroup in the adjoint group L_i^{ad} such that $T^{\text{ad}} = T/Z(L_i)$ and $B_i^{\text{ad}} \subset B_i$. Let $\mathfrak{t}^{\text{ad}} = \text{Lie } T^{\text{ad}}$ and let $\mathfrak{n}_i$ be the nilpotent part of $\mathfrak{b}_i$. Then there is a vector bundle:

$$E = L_i \times^{B_i} (\mathfrak{t}^{\text{ad}} \times \mathfrak{t} \times \mathfrak{n}_i) \rightarrow \text{St}_i^+ = L_i \times^{B_i} (\mathfrak{t} \times \mathfrak{n}_i)$$

This vector bundle admits a section

$$s : \text{St}_i^+ \rightarrow E$$

$$(g, \tau, x) \mapsto (g, \tau \bmod Z(L_i), \tau, x)$$

and St_i^{nil} is the zero-set of the section s . Thus, by part (e) of the lemma,

$$[\text{St}_i^{\text{nil}}] = \text{eu}(E) \cap [\text{St}_i^+] \in H_{T \times \mathbb{C}^\times}^2(\text{St}_i^+).$$

The line bundle $E \rightarrow \text{St}_i^+$ naturally lifts the trivial line bundle $L_i \times^{B_i} \mathfrak{t}^{\text{ad}}$, namely, the following diagram commutes:

$$\begin{array}{ccc} E & \longrightarrow & \text{St}_i^+ \\ \downarrow \pi & & \downarrow \pi \\ L_i \times^{B_i} \mathfrak{t}^{\text{ad}} & \longrightarrow & L_i/B_i \end{array}$$

Hence, $\text{eu}(E) = p^* \text{eu}(L_i \times^{B_i} \mathfrak{t}^{\text{ad}}) = u \in H_{T \times \mathbb{C}^\times}^2(L_i/B_i)$.

Therefore, $[\text{St}_i^{\text{nil}}] = u \cap [\text{St}_i^+]$ as required. $\square$

Relation **G3** holds clearly in the case $\xi = \delta$ by construction of the action of ϵ , the image of δ in $\mathfrak{H}_\nu^{\text{gr}}$.

Now suppose $\xi = \Lambda_{\text{can}}$. We have seen that there is a determinant line bundle $\det_P$ on the affine flag variety Fl_P for each parahoric subgroup P . Let

$$\Lambda_P = \Lambda_{\text{can}} + 2\rho_G - 2\rho_P + \langle \nu\rho^\vee + \partial, 2\rho_P - 2\rho_G \rangle \epsilon.$$

Then $\mathcal{L}(\Lambda_P) \cong \pi_P^* \det_P$ as $\mathbb{G}_m(\nu)$ -equivariant line bundles. The relation **G3** is linear, and since the relation holds for $\xi \in \mathfrak{a}^* \oplus \mathbb{Q}\epsilon$, by the definition of Λ_P relative to Λ_{can} it suffices to check the relation for each affine simple reflection s_i and each element $\Lambda_i := \Lambda_{P_i}$ that

$$s_i \Lambda_i - {}^{s_i} \Lambda_i s_i = \langle \Lambda_i, \alpha_i^\vee \rangle u. \quad (3.10)$$

By proposition 5.1.5 in [Yun14] seen previously,

$$\langle \Lambda_i, \alpha_i^\vee \rangle = -\deg(\mathcal{L}(\Lambda_i)|C_i) = -\deg(\pi_{P_i}^* \det_{P_i} | C_i) = 0$$

Hence ${}^{s_i} \Lambda_i = \Lambda_i$ and (3.10) becomes:

$$s_i \Lambda_i = \Lambda_i s_i.$$

By the construction of the s_i action on $q_{\nu,!}(\mathbb{Q})$, taking the cup product with a class pulled back from $\text{Sp}_{P_i, \nu}$, in particular the class $c_1^{\mathbb{G}_m(\nu)}(\det_{P_i})$, commutes with the action of s_i , and this gives the relation **G3** in this case.

Therefore the relation **G3** holds for all $\xi \in \mathfrak{a}_{\text{KM}}^*$.

3.2.4 Relation (G4)

The action of Ω_I on the affine flag variety Fl is defined by right multiplication. Denote this action by $\omega \mapsto R_\omega$. Then the natural relation that holds for all $\omega \in \Omega_I$ and $\eta \in A_{\text{KM}}$ is: $\text{Ad}(\omega^{-1})(\eta)\omega = \eta\omega$. Therefore by definition of the action of Ω_I on Fl gives

$$R_{\text{Ad}(\omega^{-1})\eta} \circ R_\omega = R_\eta \circ R_\omega$$

This relation lifts to homology, so that we have the relation

$$x_{\omega\xi} R_{\omega}^* = R_{\omega}^* x_{\xi}$$

in $H_*(\mathrm{Sp}_{\gamma})$, where $x_{\xi} = c_1(\mathcal{L}(\xi))$ for $\xi \in \mathfrak{a}_{\mathrm{KM}}^*$, as required.

3.2.5 Action of the bilinear symmetric form B_{KM} .

Each of the line bundles ξ for $\xi \in \mathfrak{a}_{\mathrm{KM}}^*$ carry an A_{KM} -equivariant structure, and letting $x(\xi) \in H_{A_{\mathrm{KM}}}^2(\mathrm{Fl})$ denote the A_{KM} -equivariant Chern class of $\mathcal{L}(\xi)$, we can extend x to a graded ring homomorphism

$$x : \mathrm{Sym}^{\bullet}(\mathfrak{a}_{\mathrm{KM}}^*) \rightarrow H_{A_{\mathrm{KM}}}^{2\bullet}(\mathrm{Fl}).$$

There is also a ring homomorphism from the cohomology of a point:

$$y : \mathrm{Sym}^{\bullet}(\mathfrak{a}_{\mathrm{KM}}^*) \cong H_{A_{\mathrm{KM}}}^{2\bullet}(\mathrm{pt}) \rightarrow H_{A_{\mathrm{KM}}}^{2\bullet}(\mathrm{Fl}).$$

It can be shown ([Yun14], lemma 5.1.6) that the image of the symmetric bilinear form B_{KM} is the same under x and y : $x(B_{\mathrm{KM}}) = y(B_{\mathrm{KM}}) \in H_{A_{\mathrm{KM}}}^4(\mathrm{Fl})$. Thus, the action of B_{KM} determined by construction (2) is the same as taking the cup product with the image of B_{KM} under the restriction $H_{A_{\mathrm{KM}}}^4(\mathrm{pt}) \rightarrow H_{\mathbb{G}_m(\nu)}^4(\mathrm{pt})$. By definition of the one-dimensional torus $\mathbb{G}_m(\nu) \hookrightarrow A_{\mathrm{KM}}$, the embedding is given by the cocharacter $d\rho^{\vee} + m\partial = m(\frac{d}{m}\rho^{\vee} + \partial) = m(\nu\rho^{\vee} + \partial)$. Therefore the action of B_{KM} on $q_{\nu,!}(\mathbb{Q})$ is given by

$$B_{\mathrm{KM}}(\nu\rho^{\vee} + \partial, \nu\rho^{\vee} + \partial)\epsilon^2 = \nu^2\mathbb{B}(\rho^{\vee}, \rho^{\vee})\epsilon^2 \in H_{\mathbb{G}_m(\nu)}^4(\mathrm{pt}).$$

Chapter 4

Further Results

We state the following results of Yun, [OY14].

4.1 The graded Cherednik algebra

The main theorem in section three has the following corollary:

Corollary. *For $a \in \mathfrak{c}(F)_\nu^{rs}$, there is a graded action of $\mathfrak{H}_\nu^{gr}/(B_{KM} - \nu^2 \mathbb{B}(\rho^\vee, \rho^\vee)\epsilon^2)$ on $H_{\mathbb{G}_m(\nu)}^*(Sp_a)$. Specialising to $\epsilon = 1$, there is an action of $\mathfrak{H}_{\nu, \epsilon=1}^{gr}$ on $H_{\epsilon=1}(Sp_\gamma)$ that is compatible with the filtrations on $\mathfrak{H}_{\nu, \epsilon=1}^{gr}$ and $H_{\epsilon=1}(Sp_\gamma)$.*

In fact, the constructions from section 3 given on the equivariant cohomology of homogeneous Springer fibers hold also for the whole flag variety:

Proposition. *There is an action of $\mathfrak{H}_\nu^{gr}/(B_{KM} - \nu^2 \mathbb{B}(\rho^\vee, \rho^\vee)\epsilon^2)$ on $H_{\mathbb{G}_m(\nu)}^*(Fl)$ such that the restriction $H_{\mathbb{G}_m(\nu)}^*(Fl) \rightarrow H_{\mathbb{G}_m(\nu)}^*(Sp_a)$ is $\mathfrak{H}_\nu^{gr}$ -equivariant for all $a \in \mathfrak{c}(F)_\nu^{rs}$.*

This allows us to obtain a geometric model for a polynomial representation of $\mathfrak{H}_\nu^{gr}$:

Lemma. *Let $\overline{\mathfrak{H}}_\nu^{gr} = \mathfrak{H}_\nu^{gr}/(B_{KM} - \nu^2 \mathbb{B}(\rho^\vee, \rho^\vee)\epsilon^2)$. There is a natural morphism of $\mathfrak{H}_\nu^{gr}$ -modules*

$$Ind_{\mathbb{Q}[\epsilon] \otimes \mathbb{Q}[W_{aff}]}^{\overline{\mathfrak{H}}_\nu^{gr}}(\mathbb{Q}[\epsilon]) \rightarrow H_{\mathbb{G}_m(\nu)}^*(Fl)$$

where the induced module is called the polynomial representation of $\overline{\mathfrak{H}}_\nu^{gr}$. Specialising to $\epsilon = 1$, this morphism becomes an isomorphism.

It is furthermore possible to analogously construct a global action of the graded Cherednik algebra on the derived image of the constant sheaf along the parabolic Hitchin fibration associated to the weighted projective line, where the weights are related to the parameter ν . Furthermore this global action is compatible with the local action constructed in section 3. [Yun11] [Yun14]

4.2 The rational Cherednik algebra

The category $\mathcal{O}$ for rational Cherednik algebras was described in [GGOR03] and consists of $\mathfrak{H}_{\nu, \epsilon=1}^{rat}$ -modules with locally nilpotent $\text{Sym}(\mathfrak{a})$ action. Given an irreducible representation of

the Weyl group W , extend the action to $\text{Sym}(\mathfrak{a}) \rtimes W$, with trivial $\text{Sym}(\mathfrak{a})$ -action. Inducing to $\mathfrak{H}_{\nu, \epsilon=1}^{\text{rat}}$, we get a module

$$\mathfrak{M}_{\nu}(\tau) = \text{Ind}_{\text{Sym}(\mathfrak{a}) \rtimes W}^{\mathfrak{H}_{\nu, \epsilon=1}^{\text{rat}}}(\tau)$$

for each irreducible representation τ of W . Let $\mathfrak{L}_{\nu}(\tau)$ be the irreducible quotient of $\mathfrak{M}_{\nu}(\tau)$ by the sum of all its proper submodules.

Then using homogeneous Hitchin fibers, it is possible to geometrically construct the irreducible spherical module $\mathfrak{L}_{\nu}(\text{triv})$.

Let Fl° be the neutral component of the flag variety of G , let ν be elliptic and $a \in \mathfrak{c}(F)_{\nu}^{\text{rs}}$.

There is a canonical algebra isomorphism

$$\mathbb{Q}[\epsilon, \epsilon^{-1}] \otimes_{\mathbb{Q}} \text{Sym}(\mathfrak{a}^*) \xrightarrow{\sim} H_{\mathbb{G}_m(\nu)}^*(\text{Fl}^{\circ})[\epsilon^{-1}] \quad (4.1)$$

There is a restriction map

$$\iota_a^* : H_{\mathbb{G}_m(\nu)}^*(\text{Fl}^{\circ}) \rightarrow H_{\mathbb{G}_m(\nu)}^*(\text{Sp}_a)^{\tilde{S}_a \rtimes B_a} \quad (4.2)$$

and this morphism is surjective after specialising $\epsilon = 1$, where $\tilde{S}_a$ is the centraliser of $\gamma = \kappa(a)$ in a certain Levi subgroup associated to the slope ν , and $B_a = \pi_1(\mathfrak{c}(F)_{\nu}^{\text{rs}}, a)$ is a certain braid group associated to ν .

Definition. The *Chern filtration* $C_{\leq i} H_{\mathbb{G}_m(\nu)}^*(\text{Fl}^{\circ})[\epsilon^{-1}]$ on $H_{\mathbb{G}_m(\nu)}^*(\text{Fl}^{\circ})[\epsilon^{-1}]$ is the image of polynomials of degree $\leq i$ in $\mathfrak{a}^*$ under the map (4.1).

The *Chern filtration* $C_{\leq i} H_{\mathbb{G}_m(\nu)}^*(\text{Fl}^{\circ})$ on $H_{\mathbb{G}_m(\nu)}^*(\text{Fl}^{\circ})$ is

$$C_{\leq i} H_{\mathbb{G}_m(\nu)}^*(\text{Fl}^{\circ}) = H_{\mathbb{G}_m(\nu)}^*(\text{Fl}^{\circ}) \cap C_{\leq i} H_{\mathbb{G}_m(\nu)}^*(\text{Fl}^{\circ})[\epsilon^{-1}].$$

The *Chern filtration* $C_{\leq i} H_{\mathbb{G}_m(\nu)}^*(\text{Sp}_a)^{\tilde{S}_a \rtimes B_a}[\epsilon^{-1}]$ on $H_{\mathbb{G}_m(\nu)}^*(\text{Sp}_a)^{\tilde{S}_a \rtimes B_a}[\epsilon^{-1}]$ is the image of $C_{\leq i} H_{\mathbb{G}_m(\nu)}^*(\text{Fl}^{\circ})[\epsilon^{-1}]$ under ι_a^* .

The *Chern filtration* $C_{\leq i} H_{\mathbb{G}_m(\nu)}^*(\text{Sp}_a)^{\tilde{S}_a \rtimes B_a}$ on $H_{\mathbb{G}_m(\nu)}^*(\text{Sp}_a)^{\tilde{S}_a \rtimes B_a}$ is

$$H_{\mathbb{G}_m(\nu)}^*(\text{Sp}_a)^{\tilde{S}_a \rtimes B_a} \cap C_{\leq i} H_{\mathbb{G}_m(\nu)}^*(\text{Sp}_a)^{\tilde{S}_a \rtimes B_a}[\epsilon^{-1}]$$

.

Then we have the following:

Proposition. Let $a \in \mathfrak{c}(F)_{\nu}^{\text{rs}}$. There is a bigraded action of $\mathfrak{H}_{\nu}^{\text{rat}}$ on $\text{Gr}_*^C H_{\mathbb{G}_m(\nu)}^*(\text{Sp}_a)^{\tilde{S}_a \rtimes B_a}$ such that the map

$$\text{Gr}_*^C H_{\mathbb{G}_m(\nu)}^*(\text{Fl}^{\circ}) \rightarrow \text{Gr}_*^C H_{\mathbb{G}_m(\nu)}^*(\text{Sp}_a)^{\tilde{S}_a \rtimes B_a}$$

is a morphism of $\mathfrak{H}_{\nu}^{\text{rat}}$ -modules, and is surjective after specialising $\epsilon = 1$.

Theorem. Let $a \in \mathfrak{c}(F)_{\nu}^{\text{rs}}$.

1. There is a geometrically defined filtration $P_{\leq} H_{\mathbb{G}_m(\nu)}^*(\text{Sp}_a)^{\tilde{S}_a}$ that is stable under the B_a -action and extends the Chern filtration on $H_{\mathbb{G}_m(\nu)}^*(\text{Sp}_a)^{\tilde{S}_a \rtimes B_a}$, such that the bigraded action of the rational Cherednik algebra $\mathfrak{H}_{\nu}^{\text{rat}}$ on $\text{Gr}_*^C H_{\mathbb{G}_m(\nu)}^*(\text{Sp}_a)^{\tilde{S}_a \rtimes B_a}$ extends to $\text{Gr}_*^P H_{\mathbb{G}_m(\nu)}^*(\text{Sp}_a)^{\tilde{S}_a}$ and commutes with the B_a -action.
2. Specialising to $\epsilon = 1$, the $\mathfrak{H}_{\nu, \epsilon=1}^{\text{rat}}$ -module $\text{Gr}_*^C H_{\mathbb{G}_m(\nu)}^*(\text{Sp}_a)^{\tilde{S}_a \rtimes B_a}$ is isomorphic to the irreducible finite-dimensional spherical module $\mathcal{L}(\text{triv})$ of $\mathfrak{H}_{\nu, \epsilon=1}^{\text{rat}}$.

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